

E1733A Automation Interface Operation

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E1733A Automation Interface Operation

General Operation of the Automation Interface

Introduction

E1733A Dimensional Metrology Software (Rev. 1.11 and later) can be remotely controlled by a user's custom program, adjusting window appearance, setting up/starting/stopping measurements, changing configurations, collecting data, printing reports, etcetera, all without clicking buttons or manually typing on the keyboard. This functionality allows one to automate the machine calibration process, reducing labor costs and improving work efficiency.

An alternative way to implement measurement automation is to use the Application Programming Interface (API) to the hardware. The main difference between the automation interface and the API is that the API is a group of functions, packed in a Dynamic Link Library (DLL). The programmer must understand how both the hardware and all these functions run, then call them with proper parameters in the correct sequence, and write the code to handle all possible exceptions, as well as create a user interface. This requires a skilled programmer and some potentially complicated coding.

What E1733A Automation interface offers is a way to programmatically control all the functions of the existing E1733A calibration software's graphical user interface (GUI). With E1733A Automation, the user doesn't need to know anything about the API functions or hardware operating details. They just need to know how to use E1733A software itself, and then access and control it with their program, instead of a keyboard and mouse.

Another difference is that the user can still access E1733A software manually when E1733A Automation is working (although this can optionally be disabled as well). This functionality makes it possible to run the E1733A software in mixed mode, controlled by an automation program and a user simultaneously. When using the API, it is up to the programmer to create any needed user interface, and then handle all the input and output associated with that interface.

A basic automation program can be created with any computer language capable of calling Windows message functions. The automation program sends messages to the E1733A software and processes the returned data. All that is needed is a general programming knowledge and how to call a few Windows message related functions.

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How to use the Automation Commands

The following paragraphs illustrate what is needed to start, link to, send messages to, and pass additional information to/from, the E1733A software running in automation mode.

Starting E1733A Automation

To run the E1733A software in automation mode, the user must run E1733A.exe with a command line parameter:

```
C:\><Program Path>\E1733A /RemoteName=<User Defined Name>
```

where <Program Path> is the path of E1733A.exe, and <User Defined Name> can be any text that has no space nor quotation mark. This can be done by calling the *ShellExecute()* function in the user's program. For example:

```
.....
#define E1733APath "C:\\Program Files\\Keysight\\E1733A\\E1733A.exe"
#define Parameter "/RemoteName=MyTestAuto"
.....
if (ShellExecute(0, NULL, E1733APath, Parameter, NULL, SW_SHOW) == NULL)
.....
```

When E1733A is started, <User Defined Name> will be shown in its title, which helps the user locate the window handle for E1733A software in the next step.

Getting the handle

In order to send messages to the E1733A software, one must obtain the window handle for the E1733A window. Below is a sample C program routine for this task. Three Windows API functions are used: *FindWindow()*, *GetWindowText()* and *GetWindow()*.

```
HWND SearchE1733A(char* Caption)
{
    char Buffer[BufferSize];
    HWND TheHandle = FindWindow(0, 0);
    while (TheHandle != 0)
    {
        GetWindowText(TheHandle, Buffer, BufferSize);
        if (strcmp(E1733ATitle, Buffer) == 0)
        {
            return TheHandle;
        }
        else
        {
            TheHandle = GetWindow(TheHandle, GW_HWNDNEXT);
        }
    }
    return 0;
}
```

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For proper operation, the <User Defined Name> created when starting the program should be enclosed in angle brackets ('<' and '>'), prefixed with "Keysight E1733A ", and passed to this function as the parameter (e.g. "Keysight E1733A <MyTestAuto>"). Make sure that the name is unique among all windows and does not contain any spaces or quotation marks.

Note that the user's automation program must wait a few seconds after starting the E1733A software before calling this search function. The delay allows the E1733A software to complete its startup process and become visible to the search routine.

```
HWND E1733A;  
UINT Try;  
.....  
Try = 0;  
do // repeat until E1733A is started and its handle is acquired successfully  
{  
    Sleep(100);  
    Try++;  
    E1733A = SearchE1733A(E1733ATitle);  
}
```

Sending messages

One must call the window function *SendMessage()* to send a message to the E1733A software. This Windows API routine, with four 32-bit parameters and a 32-bit return value, is defined as follows:

```
LRESULT SendMessage( HWND hWnd, UINT Msg, WPARAM wParam, LPARAM lParam);
```

The first parameter is the handle of the E1733A window, which can be obtained using the above sample *SearchE1733A()* routine, or any other method that provides the proper handle.

The second parameter is a 32-bit integer that represent the message for the E1733A software. In this case, it is always *E1733A_WM_REMOTECOMMAND*, which is defined as:

```
#define E1733A_WM_REMOTECOMMAND 0x0500
```

The third and fourth parameters, which specify the message, are defined below. They tell the E1733A software what to do. The information required for all messages, with each message triggering only one action, is packed into the following two unions, and sent to the function as these last two parameters.

```
typedef union {  
    LONG AsWParam; // basic parameter of the message  
    struct {  
        unsigned char Code; // indicating type of action, defined as E1733A_CC_xxxx  
        unsigned char Index; // target or action sub-type, defined as E1733A_CI_xxxx  
        unsigned short Num; // index num of items or run num of the pos error data  
    } Command;  
} TWParam;
```

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```
typedef union {  
    LONG AsLParam; // additional parameter of the message  
    struct {  
        signed short X; // X position or width  
        signed short Y; // Y position or height  
    } XY;  
    LONG Pos; // position number of the error data  
} TLParam;
```

For the first parameter, set:

- Command.Code to a *E1733A_CC_xxxx* constant to initiate the desired action.
- Command.Index to the desired *E1733A_CI_xxxx* constant or zero, depending on the action.
- Command.Num to a value or zero, depending on the type of action.

For the second parameter, the *XY.X*, *XY.Y*, and *Pos* structure elements are set to a value or to zero, depending on the type of action.

Below is a sample for moving the window position of E1733A software by sending it a message.

```
UINT Result;  
TWParam WP;  
TLParam LP;  
.....  
WP.Command.Code = E1733A_CC_SETE1733ASTATUS;  
WP.Command.Index = E1733A_CI_MOVETO;  
WP.Command.Num = 0;  
LP.XY.X = 300;  
LP.XY.Y = 10;  
Result = SendMessage(E1733A, E1733A_WM_REMOTECOMMAND,  
    WP.AsWParam, LP.AsLParam);
```

Note that all unused variables should be set to zero. See the next section for a mapping of all on screen elements to specific *E1733A_CC_xxxx* and *E1733A_CI_xxxx* constants. Also see the *E1733A_Remote.h* file for details on setting the numeric elements of the *TWParam* and *TLParam* structures.

The result of each action is returned via the 32-bit return value of *SendMessage()*. For most actions, it is an error code that is defined by the *E1733A_RC_xxxx* constants (see the *E1733A_Remote.h* file for a listing of these constants). Some actions may return an integer number, a float value, or a boolean value. To handle these situations, the return value of *SendMessage()* should be converted to the proper data type using the union defined below.

```
typedef union {  
    LONG Integer; // result is integer, boolean, or index defined by E1733A_RC_xxxx  
    float Single; // result is single precision floating point number  
} TIntFloat;
```

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For example, to read the position error data for axis 1, run 2, position 3, use TIntFloat as follows:

```
TIntFloat Res;  
.....  
WP.Command.Code = E1733A_CC_READERRORDATA;  
WP.Command.Index = E1733A_CI_ERRAXIS1; // axis number = 1  
WP.Command.Num = 2; // run number = 2  
LP.Pos = 3; // position number = 3  
Res.Integer = SendMessage(E1733A, E1733A_WM_REMOTECOMMAND,  
                          WP.AsWParam, LP.AsLParam);  
printf("Position Error Value = %5.3f\n", Res.Single);
```

If the required floating point data is not available, NAN (not a number) will be returned. For boolean return values, use the .Integer portion of the structure, where 0 stands for false and 1 stands for true.

Reading and writing strings

Some actions require strings to be passed between the E1733A software and the user's automation program. In such cases, the 32-bit return value from *SendMessage()* is insufficient, and a file mapping technique is used to transfer more information.

The first step requires, prior to starting the E1733A software, that the user's automation program create a mapped file by calling the Windows API function *CreateFileMapping()*. This file will exist in memory and gets mapped to both the E1733A software and the user's automation program. It thus provides a mutually accessible memory area where strings can be passed between the two programs. When creating the mapped file, the <User Defined Name>, which gets passed to E1733A software as its command line parameter, must exactly match the last parameter, which specifies the mapped file's name. If they do not match, then the E1733A software will not be able to access to the mapped file.

```
#define MemFileSize 1024  
.....  
HANDLE MappedFile;  
.....  
MappedFile = CreateFileMapping(INVALID_HANDLE_VALUE, NULL,  
                              PAGE_READWRITE, 0, MemFileSize, "MyTestAuto");  
.....
```

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After creating the mapped file, the user's automation program can read strings from, or write strings to, the E1733A software by using the pointer returned from calling the Windows API *MapViewOfFile()* function as shown in the below sample for reading the operator's name. Make sure that the mapped file size is large enough to contain the string that is to be read.

```
char* Ptr;  
  
.....  
WP.Command.Code = E1733A_CC_READSETUPVALUE;  
WP.Command.Index = E1733A_CI_INFSETUP_OPERATOR_ITEMTEXT;  
WP.Command.Num = MemFileSize; // if size is not enough, returned string will be truncated  
LP.AsLParam = 0;  
Result = SendMessage(E1733A, E1733A_WM_REMOTECOMMAND,  
                        WP.AsWParam, LP.AsLParam);  
Ptr = (char*)MapViewOfFile(MappedFile, FILE_MAP_WRITE, 0, 0, WP.Command.Num);  
  
.....  
if (Ptr != NULL) UnmapViewOfFile(Ptr);
```

Note that to avoid a memory leak, the Windows API *UnmapViewOfFile()* function must be called when the pointer is no longer used. The handle of the mapped file must also be released by calling the *CloseHandle()* function when it is no longer used.

```
CloseHandle(MappedFile);
```

Please visit Microsoft's web site to get more help for the Windows API functions mentioned above.

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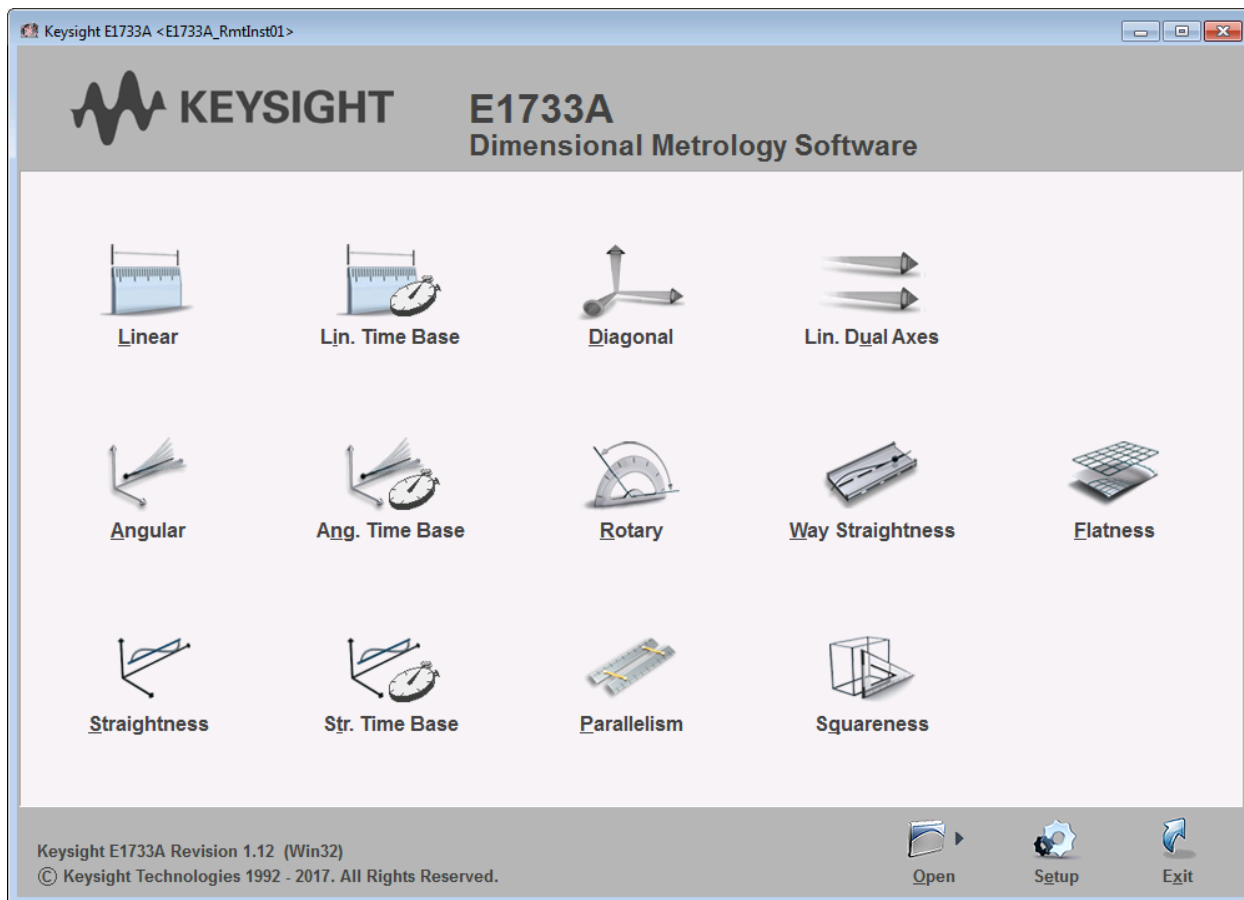
Automation Commands

The following sections contain screen captures and tables of *E1733A_CC_xxxx* and *E1733A_CI_xxxx* constants. Generally there is a one to one, positional correspondence, between the screen elements and the constants in the table. In any ambiguous situations, the constant names can be used to help clarify the action that will occur.

New Command ('CC_NEW¹)

Use the 'CI_*' constants in the table with the 'CC_NEW' command to open a measurement dialog of the desired type.

Main Menu:

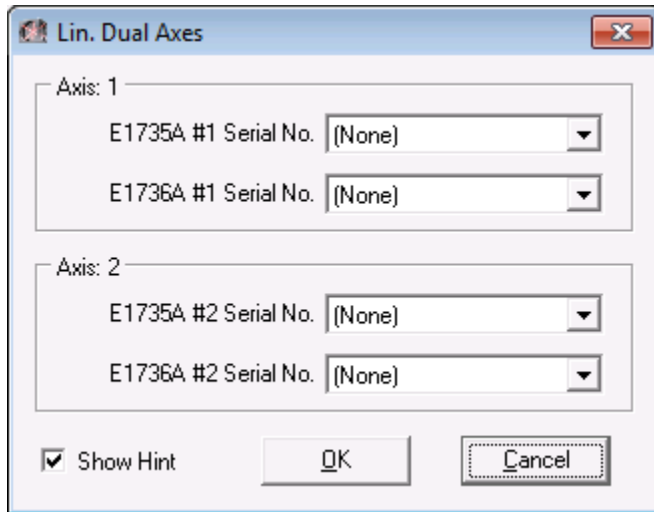


'CI_LINEAR	'CI_LINTIMEBASE	'CI_DIAGONAL	'CI_LINDUALAXES	
'CI_ANGULAR	'CI_ANGTIMEBASE	'CI_ROTARY	'CI_WAYSTRAIGHTNESS	'CI_FLATNESS
'CI_STRAIGHTNESS	'CI_STRTIMEBASE	'CI_PARALLELISM	'CI_SQUARENESS	

¹ There may or may not be an apostrophe before any CC or CI constant to indicate that a portion of the actual constant has been omitted to make reading thru this manual easier. The portion omitted is as indicated in a table's 1st row, or "E1733A_", or "E1733A_CC_", or "E1733A_CI_" as appropriate (also indicated in the table's 1st row).

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Lin Dual Axis:



The above dialog is shown in response to the 'CC_NEW with 'CI_LINDUALAXES message. The following message parameters are used with the 'CC_WRITESETUPVALUE command to specify or select the hardware for each axis as well as read back (using the CC_READSETUPVALUE command) the associated serial number or device index value. Then, to open the two measurement dialogs, send the 'CC_EXIT with 'CI_SETUP message. Note that once this message is sent, the initial E1733A window will close and two new measurement windows will open. The user's automation program should wait a few seconds and then obtain handles for these two new windows, which will have titles with " - Linear (Axis:<AxisNumber>)" appended to the base window's title (<AxisNumber> will be either a '1' or a '2'). Finally, after completing the measurement and closing the two windows, the base window will re-open, and the automation program must obtain a new handle to access it again.

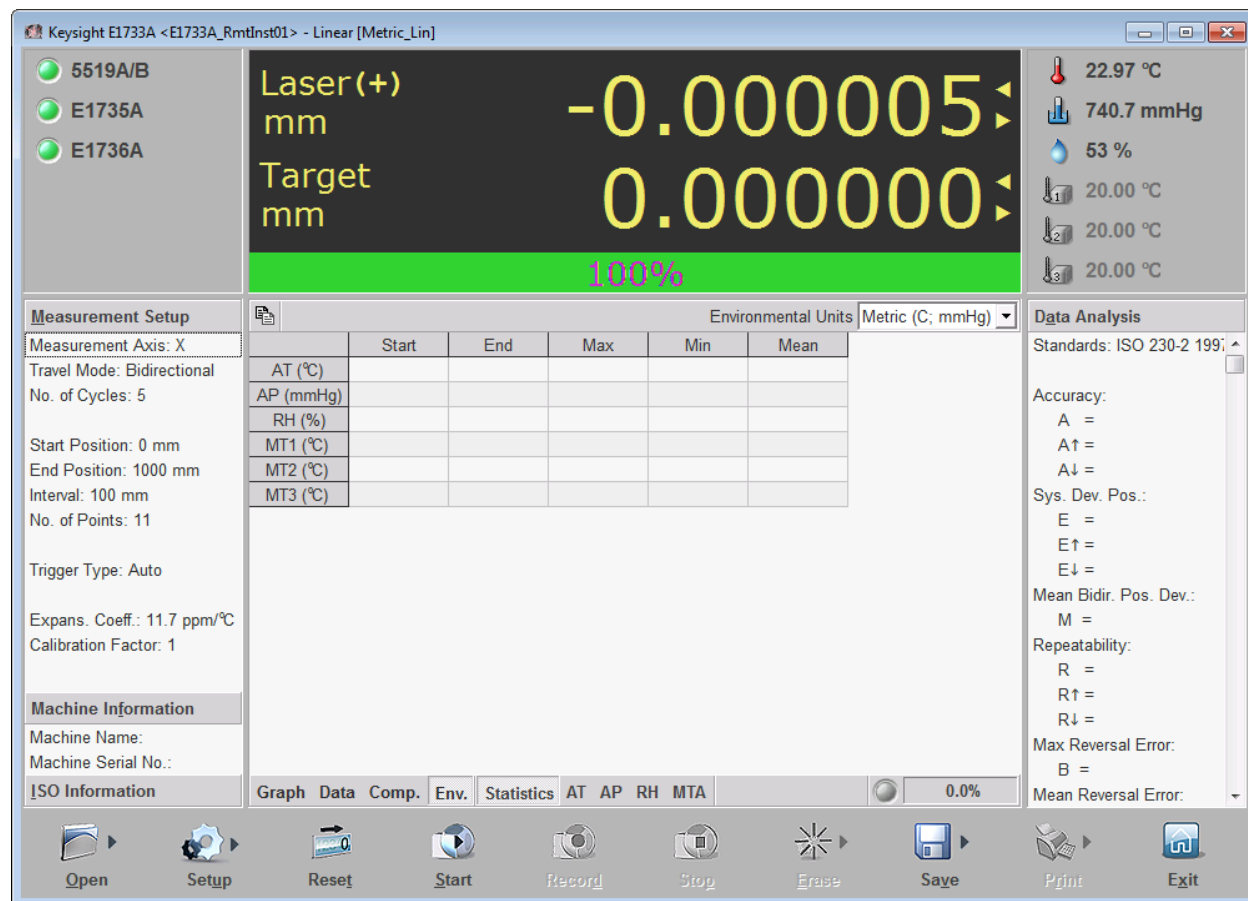
Item	'CI_LDASETUP_* Constant	What to put in shared memory
E1735A #1 Serial No.	'LDACARD1_ITEMTEXT	The module's serial number, including the "CN" prefix.
E1736A #1 Serial No.	'LDAHUB1_ITEMTEXT	
E1735A #2 Serial No.	'LDACARD2_ITEMTEXT	
E1736A #2 Serial No.	'LDAHUB2_ITEMTEXT	
E1735A #1 Serial No.	'LDACARD1_CHOOSING	0 for (None), no device
E1736A #1 Serial No.	'LDAHUB1_CHOOSING	1 for the first listed device
E1735A #2 Serial No.	'LDACARD2_CHOOSING	2 for the second listed device
E1736A #2 Serial No.	'LDAHUB2_CHOOSING	etc.
Show Hint Checkbox	E1733A_CI_TOPSETUP_SHOWHINT_CHOOSING	0 for unchecked 1 for checked
OK Button	E1733A_CC_EXIT with E1733A_CI_SETUP	No entry needed in shared memory
Cancel Button	No Command Available	--

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Operation Commands

From each measurement type's data collection screen, all functions can be accessed with various 'CC_* commands and associated 'CI_* index values as indicated with the following figures and tables.

Tabs and Buttons:



Tab	'CC_* Command	Available 'CI_* Index Values	Notes
Graph	SHOWGRAPH	GRAALL, GRAAXIS1 ... GRAAXIS8, GRAPOS, GRAVEL, GRAACC, GRAFFT	Select appropriate index constants for measurement mode in use.
Data	SHOWRAWDATATABLE	DATALL, DATAxis1 ... DATAxis8	
Comp	SHOWCOMPENSATION	COMPALL, COMPAXIS1, COMPAXIS2	'AXIS2 for merged data only
Env	SHOWENVIRONMENT	See below	
Statistics	SHOWENVIRONMENT	'CI_STAT	
AT	SHOWENVIRONMENT	'CI_AT	
AP	SHOWENVIRONMENT	'CI_AP	
RH	SHOWENVIRONMENT	'CI_RH	
MTA	SHOWENVIRONMENT	'CI_MAT	

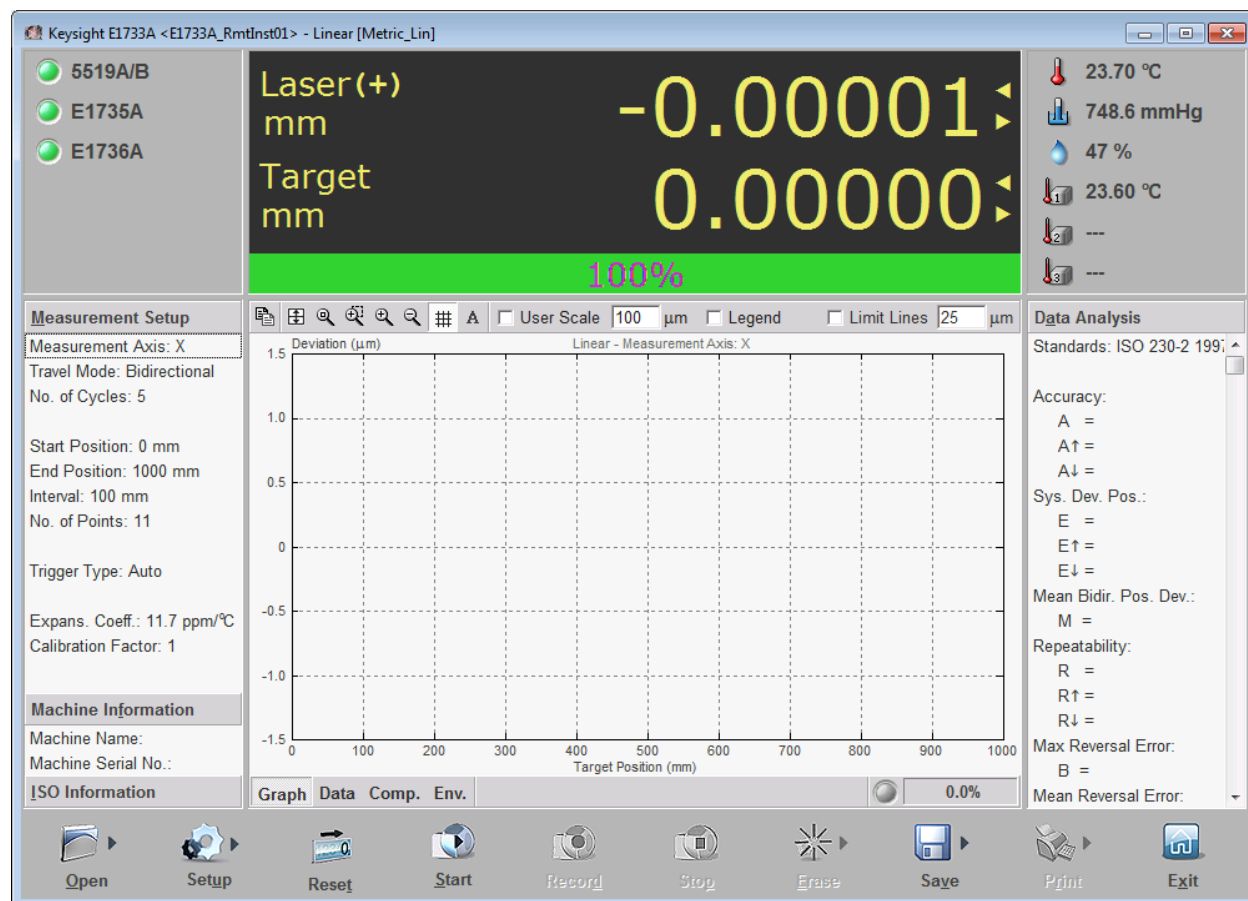
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Button	'CC_* Command	Available 'CI_* Index Values	Notes
Open	'CC_OPEN	SETUPONLY, SETUPDATA, INFOONLY, MERGEDATA, COMBINED*, COMBINEDPAR*, COMBINEDSQU*	All windows Measurement Windows only Main Window only
Setup	'CC_SETUP	See Setup Commands section.	
Reset	'CC_RESET	0	No other index values.
Start	'CC_START	0	No other index values.
Record	'CC_RECORD	0	No other index values.
Stop	'CC_STOP	0	No other index values.
Erase	'CC_ERASE	ALL, RUN, ONE, MER	MER if for Merged Data
Save	'CC_SAVE	STEUPDATA, RAWDATA_TXT, RAWDATA_CSV, RAWDATA_POS, RAWDATA_RUN, COMPTABLE_TXT, COMPTABLE_CSV, COMPTABLE_POS, COMPTABLE_RUN, ENVDATA_TXT, ENVDATA_CSV	Miss-spelling is intentional to avoid duplicate CI constants with different values.
Print	'CC_PRINT	REPORT, RAWBYPOS, RAWBYRUN, ISO1988, ISO1997, ISO2006, GB2000, E1733A_CI_ISO2014	
Exit	'CC_EXIT	MAIN, DATA, SETUP	Main closes the program. Any setup dialog should be exited prior to exiting a data dialog.

* use a comma to separate the two file names for opening a COMBINED measurement type.

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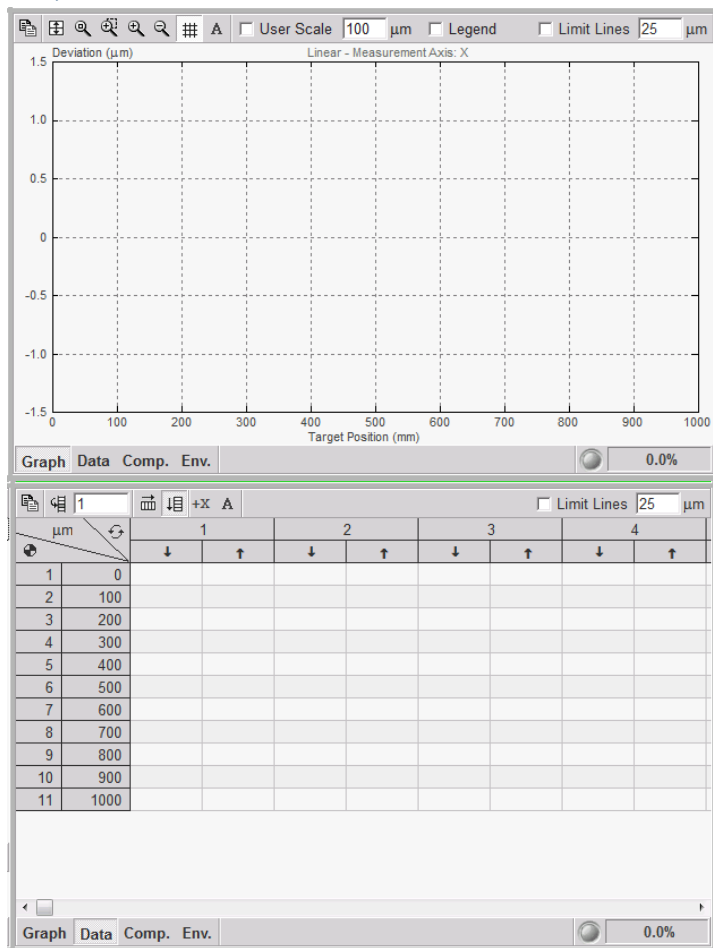
Screen controls:



Screen Control Item	'CC_* Command	'CI_* Index Value	Notes
Enable/Disable user clicks	SETE1733A STATUS	ENABLED, DISABLED	
Change window condition		SHOW, HIDE, MAXIMIZE, MINIMIZE, RESTORE	
Change window size and location		RESIZE, MOVETO, MOVESETUP	See header for how to set X and Y value.
Show Number & Graph/Data	CHANGEDISPLAY	NORMALDISPLAY	
"Maximize" Numeric		ENLARGENUMERIC	
"Maximize" Graph/Data		ENLARGEGRAPH	
Adjust number of digits	CHANGENUMERIC	INCUPPER, DECUPPER, INCLOWER, DECLOWER	
Adjust direction sense		POSUPPER, NEGUPPER, POSLOWER, NEGLOWER	
Change what is displayed in upper and lower displays	SELECTUPPERDISP	LASER, ERROR, ANGLE, VELOCITY	Active measurement mode determines available index
	SELECTLOWERDISP	TARGET, ENCODER, TIME, STATION	

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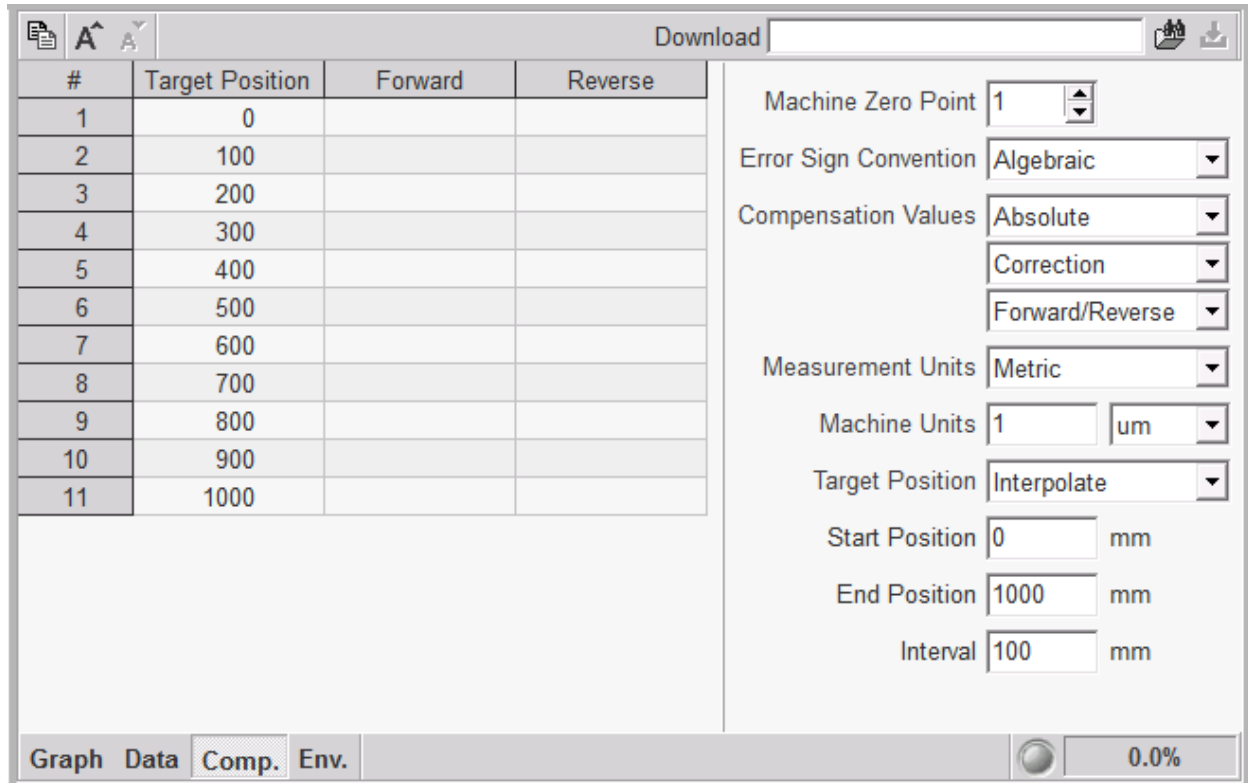
Graph and Data controls:



Graph and/or Data Control Item	'CC_* Command	'CI_DATASETUP_* Index Value	Notes
Copy control	--	--	Not available
Pan / Zoom Controls	--	--	Not available
Select First Line Cell	--	--	Not available
Change Selected Cell	--	--	Not available
Grid Lines Enable	READSETUPVALUES, WRITESETUPVALUES	GRIDLINES_CHOOSING	Set shared memory buffer text to 0 / 1 for Disable / Enable
User Scale Enable		USERSEL_CHOOSING	
Legend Enable		LEGEND_CHOOSING	
Limit Line Enable		LIMITSEL_CHOOSING	
User Scale Size		USERSCALE_ITEMTEXT	Set shared memory buffer text to desired setting
Limit Line Location		LIMITLINE_ITEMTEXT	
First Line Row Select		FIRSTROW_ITEMTEXT	
Row / Column Select		TARINROW_CHOOSING	Use 0/1 for row / col select
Error Sign ( , )		ERRORSIGN_CHOOSING	Use 0/1 for + / - sign
Error Sign Convention		E1733A_CI_CMPSETUP_C OMPABS_CHOOSING	0 for Algebraic 1 for Calibrator
3D Graph Select (flatness graph only)		FLAGRASEL_CHOOSING	Use 0/1 for 2D numbers / 3D graph

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Compensation Table controls:



#	Target Position	Forward	Reverse
1	0		
2	100		
3	200		
4	300		
5	400		
6	500		
7	600		
8	700		
9	800		
10	900		
11	1000		

Machine Zero Point: 1

Error Sign Convention: Algebraic

Compensation Values: Absolute, Correction, Forward/Reverse

Measurement Units: Metric

Machine Units: 1 um

Target Position: Interpolate

Start Position: 0 mm

End Position: 1000 mm

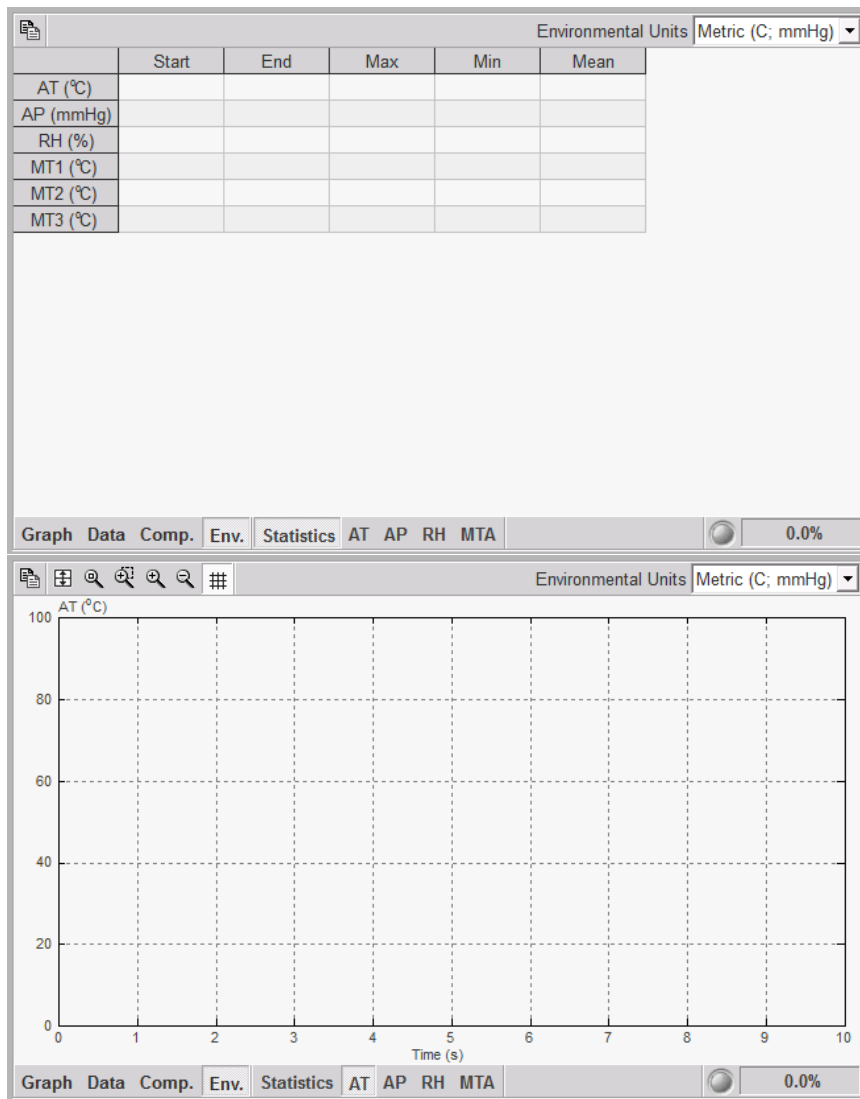
Interval: 100 mm

Graph Data **Comp.** Env. 0.0%

Compensation Table Control Item	'CC_* Command	'CI_CMPSETUP_* Index Value	Notes
Copy control	--	--	Not available
Font size controls	--	--	Not available
Browse for Program	--	--	Not available
Run Program	--	--	Not available (use other ways to run a download program)
Program Selected	READSETUPVALUES, WRITESETUPVALUES	COMPPROG_ITEMTEXT	Include full path
Machine Zero Point		'COMPZERO_ITEMTEXT	→ 'ANASETUP_* constant
Error Sign Convention		COMPABS_CHOOSING	0 / 1 for Algebraic / Calibrator
Comp Values		COMPALC_CHOOSING	0 / 1 for Absolute / Increment
Comp Type		COMPSIGN_CHOOSING	0 / 1 for Correction / Error
Directional/Combined		COMPDIR_CHOOSING	0 / 1 for Combined / Fwd:Rev
Measurement Units		UNITSYSEL_CHOOSING	0 / 1 for Metric / English
Machine Units: Value		MACHUNIT_ITEMTEXT	Put value as string in buffer
Machine Units: Units		'ERRUNIT_CHOOSING	→ 'MEASETUP_* constant
Target Position		COMPSEL_CHOOSING	0 / 1 for Trgt List / Interpolate
Start Position		COMPSTART_ITEMTEXT	Put value as string in buffer
End Position		COMPEND_ITEMTEXT	
Interval		COMPINTVL_ITEMTEXT	

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Environment controls:



Environment Control Item	'CC_* Command	'CI_ENVSETUP_* Index Value	Notes
Copy control	--	--	Not available
Pan and Zoom controls	--	--	Not available
Grid Lines Enable	READSETUPVALUES, WRITESETUPVALUES	E1733A_CI_DATASETUP_ GRIDLINES_CHOOSING	Set shared memory buffer text to 0 – Disabled, 1 – Enabled
Units Select		ENVUNITSEL_CHOOSING	Set shared memory buffer text to 0 / 1 / 2 for Metric / English / SI unit selection.

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Setup Commands ('CC_SETUP)

Use the 'CI_* constants in the sub-headings below (summarized in following table) with the 'CC_SETUP command to show the desired setup dialog. Then use the CC_READSETUPVALUE and 'CC_WRITESETUPVALUE commands with the 'CI_* constants indicated in the tables within each sub-heading to access the corresponding parameters.

Use the 'CC_EXIT command with the 'CI_SETUP value to press the "OK" button and close the dialog. However, there is no command associated with pressing the "Cancel" button.

Note that the Setup values can be accessed directly, without showing the setup dialog.

Title of Setup Dialog	'CI_* Constant
E1733A	PROGRAM
System	SYSTEM
Environmental Compensation	ENVIRONMENT
Measurement Setup	MEASUREMENT
Data Analysis	ANALYSIS
Machine Information	MACHINEINFO
ISO Information	ISOINFO

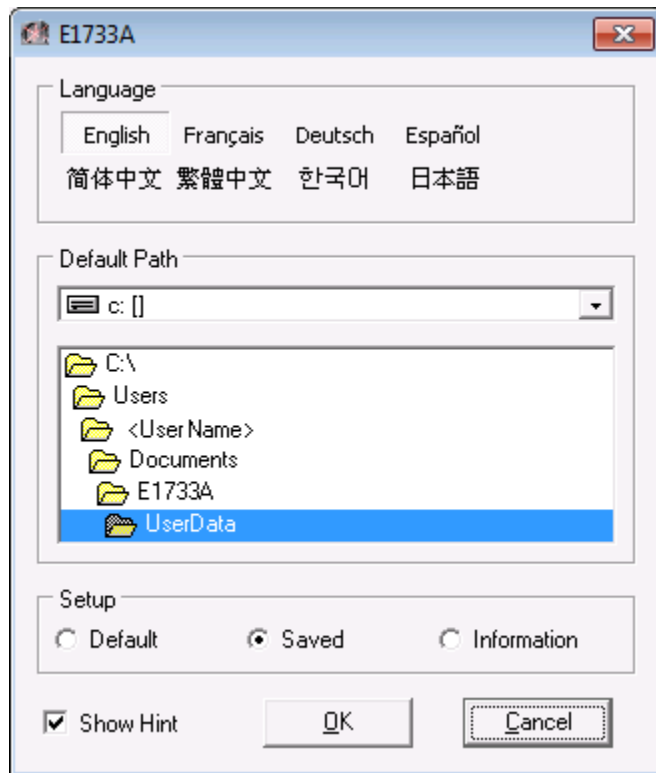
Button Commands: The rotary measurement system setup dialog (see page 18) contains some buttons that are used for calibrating the 55290B rotary table. The following constants are used with the 'CC_SETUP command to "press" these buttons. Note that attempting to "press" one of these buttons when it is disabled will return the E1733A_RC_FUNCTIONDISABLED error code as the result of the message.

Title of Button in Setup Dialog	'CI_* Constant
(+) or (-)	TOGGLELASER
Step 1: Reset laser and rotary at 0 arcdeg	CALISTEP1
Step 2: Sample between -3 and -4 arcdeg	CALISTEP2
Step 3: Sample between +3 and +4 arcdeg	CALISTEP3
Apply: User Calibration => Calibration Factor	APPLYCALI

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E1733A Setup ('CI_PROGRAM):

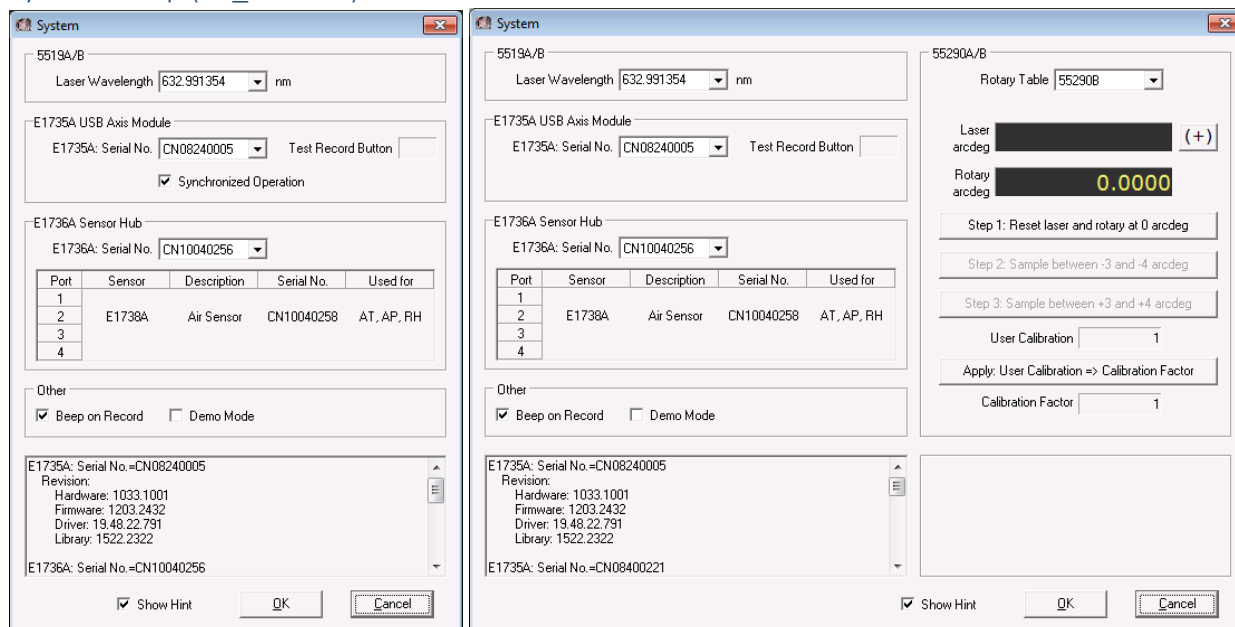
This index value only works when the Main Menu is being displayed.



Item	'CI_TOPSETUP_* Constant
Language	LANGUAGE_CHOOSING
Default Path	DATAPATH_ITEMTEXT
Setup Style	OPENSTYLE_CHOOSING
Show Hint	SHOWHINT_CHOOSING
OK	E1733A_CC_EXIT with E1733A_CI_SETUP
Cancel	No Command Available

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System Setup ('CI_SYSTEM):



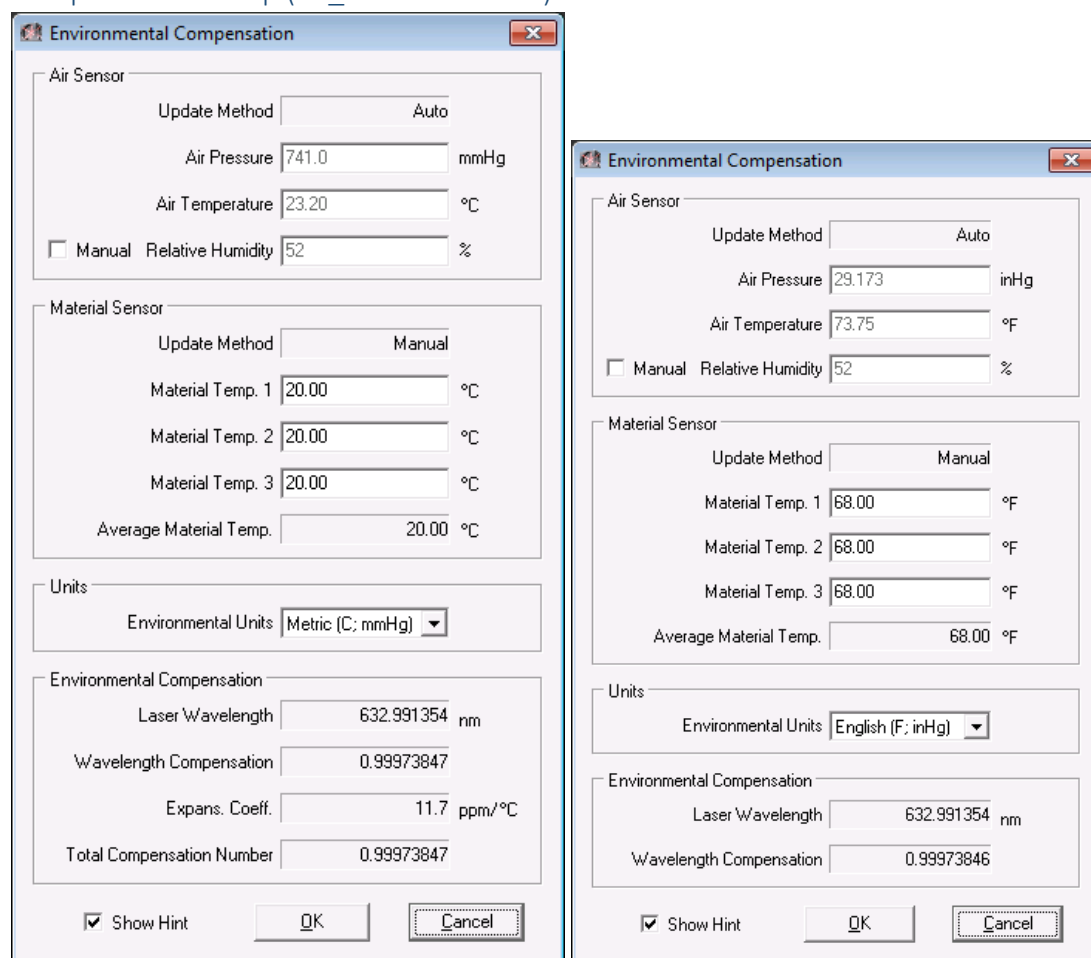
Left Column Item	'CI_SYSSETUP_* Constant	Right Column Item	'CI_SYSSETUP_* Constant
Wavelength	WAVELENGTH_ITEMTEXT	Rotary Table Type	ROTARYDEV_CHOOSING
E1735A Serial #	E1735ASN_ITEMTEXT	Direction Setting [‡]	'CI_TOGGLELASER
E1735A Serial #	E1735ASN_CHOOSING	Step 1 button [‡]	'CI_CALISTEP1
Synchronized	SYNCHOP_CHOOSING	Step 2 button [‡]	'CI_CALISTEP2
E1736A Serial #	E1736ASN_ITEMTEXT	Step 3 button [‡]	'CI_CALISTEP3
E1736A Serial #	E1736ASN_CHOOSING	Calibration Factor	USERCALI_ITEMTEXT
Beep on Record	BEEPREC_CHOOSING	Apply Button [‡]	'CI_APPLYCALI
Demo Mode	DEMOMODE_CHOOSING		
Show Hint (chk box)	E1733A_CI_TOPSETUP_SHOWHINT_CHOOSING		
OK (button)	E1733A_CC_EXIT with E1733A_CI_SETUP		
Cancel (button)	No Command Available		

[‡] The functions associated with these buttons are accessed with the 'CC_SETUP command, not the 'CC_READSETUPVALUE and 'CC_WRITESETUPVALUE commands (see the Button Commands paragraph in the Setup Commands ('CC_SETUP) section on page 16).

Note that there are two commands that can be used with each of the serial number fields. The 'ITEMTEXT commands allow specifying the serial number while the 'CHOOSING commands allow use of an index value. Having both options is useful when there are more than one E1735A or E1736A modules and the serial numbers may not be known ahead of time. One can specify an index value and then read back the serial number.

E1733A Automation Interface Operation

Compensation Setup ('CI_ENVIRONMENT):



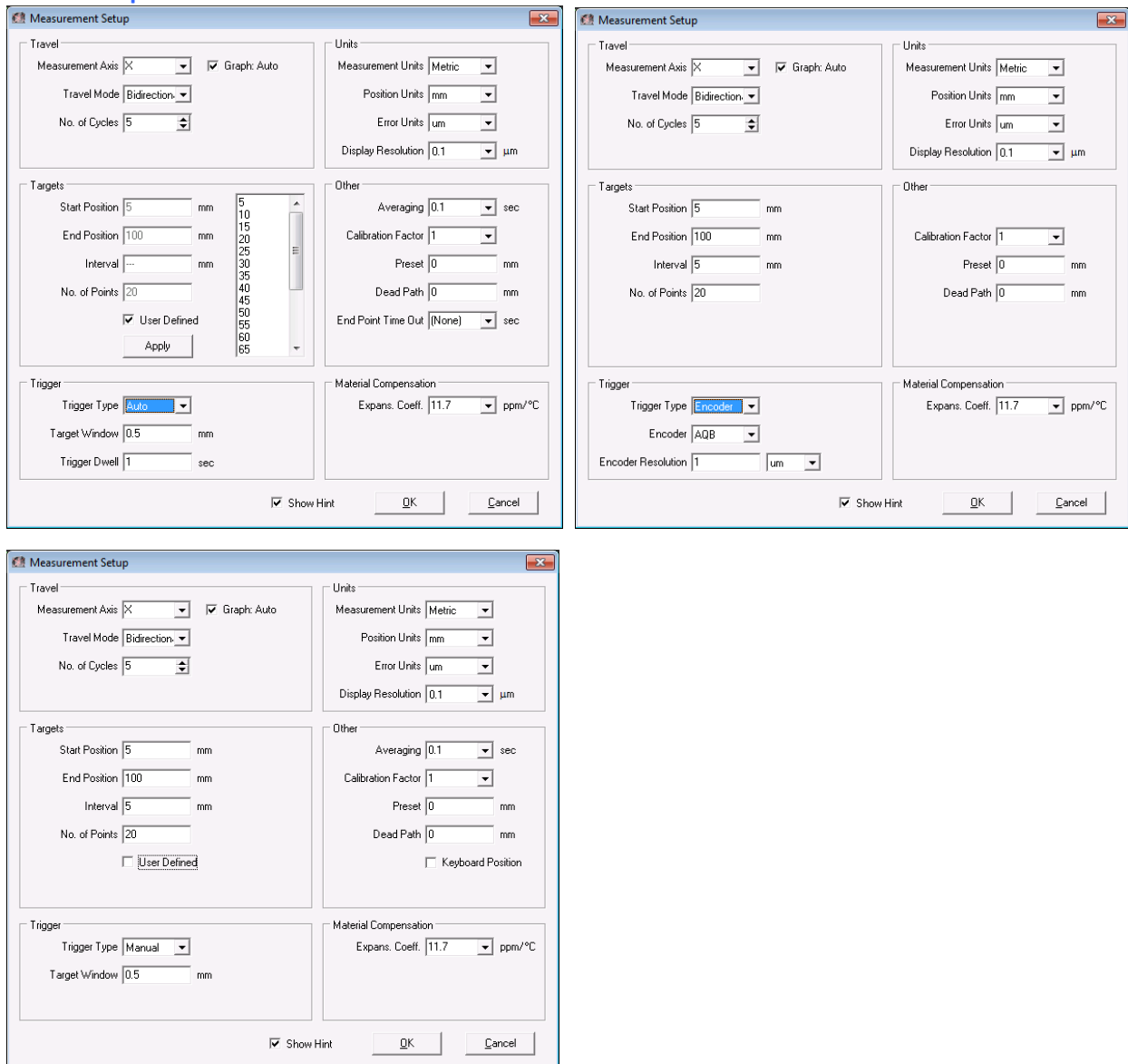
Item	'CI_ENVSETUP_* Constant	What to put in shared memory
Air Pressure Value	AIRPRES_ITEMTEXT	Desired value as text string
Air Temperature Value	AIRTEMP_ITEMTEXT	
Relative Humidity Value	RELHUMI_ITEMTEXT	
Relative Humidity Manual Selection	MANUALRH_CHOOSING	0 / 1 for Unchecked / Checked
Material Temp 1 Value	MATTEMP1_ITEMTEXT	Desired value as text string
Material Temp 2 Value	MATTEMP2_ITEMTEXT	
Material Temp 3 Value	MATTEMP3_ITEMTEXT	
Units Selection	ENVUNITSEL_CHOOSING	0 / 1 / 2 for Metric / English / SI
Laser Wavelength	E1733A_CI_SYSSETUP_WAVELENGTH_ITEMTEXT	Desired value as text string
Expans. Coeff.	E1733A_CI_MEASETUP_EXPCOEF_ITEMTEXT	
Show Hint (chk box)	E1733A_CI_TOPSETUP_SHOWHINT_CHOOSING	0 / 1 for Unchecked / Checked
OK (button)	E1733A_CC_EXIT with E1733A_CI_SETUP	--
Cancel (button)	No Command Available	--

E1733A Automation Interface Operation

Measurement Setups ('CI_MEASUREMENT'):

The appropriate measurement setup dialog will be displayed in response to this index constant depending on which measurement mode is currently open.

Linear Setup:



The following table summarizes the settings shown in the three screenshots of the Measurement Setup dialog box for the Linear Setup mode.

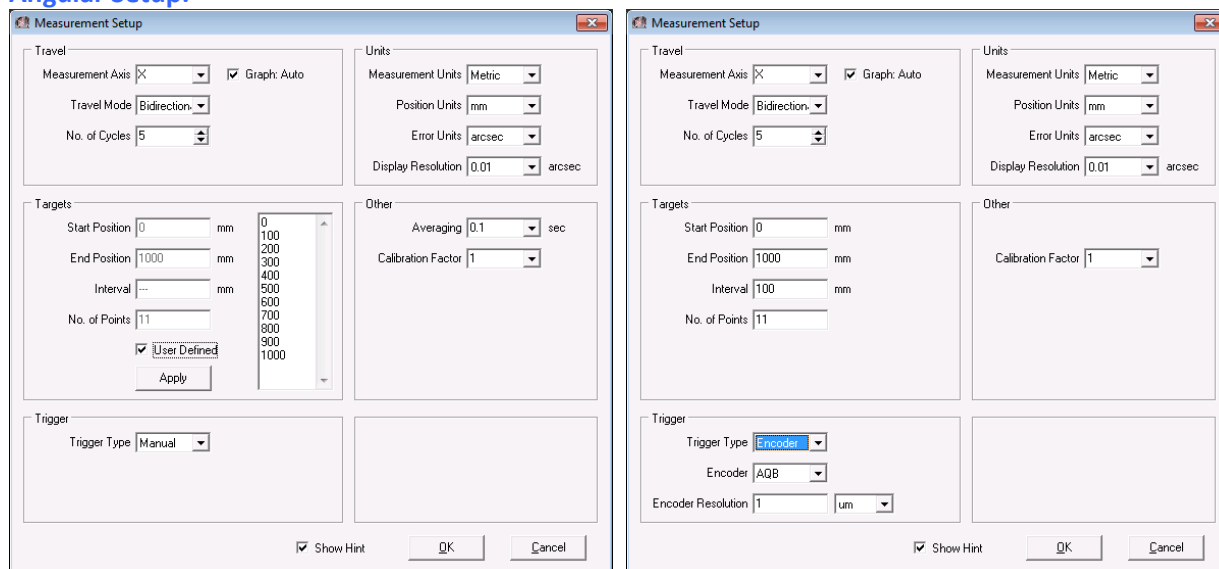
Section	Parameter	Screenshot 1 (Top Left)	Screenshot 2 (Top Right)	Screenshot 3 (Bottom)
Travel	Measurement Axis	X	X	X
	Travel Mode	Bidirection	Bidirection	Bidirection
	No. of Cycles	5	5	5
	Graph: Auto	☒	☒	☒
Units	Measurement Units	Metric	Metric	Metric
	Position Units	mm	mm	mm
	Error Units	um	um	um
	Display Resolution	0.1 um	0.1 um	0.1 um
Targets	Start Position	5 mm	5 mm	5 mm
	End Position	100 mm	100 mm	100 mm
	Interval	---	5 mm	5 mm
	No. of Points	20	20	20
	User Defined	☒	☐	☐
	Apply	[Button]	[Button]	[Button]
	Other	[List: 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65]	[List: 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65]	[List: 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65]
	Keyboard Position	☐	☐	☐
Other	Averaging	0.1 sec	0.1 sec	0.1 sec
	Calibration Factor	1	1	1
	Preset	0 mm	0 mm	0 mm
Dead Path	Dead Path	0 mm	0 mm	0 mm
	End Point Time Out	(None) sec	(None) sec	(None) sec
	Material Compensation	Expans. Coeff. 11.7 ppm/°C	Expans. Coeff. 11.7 ppm/°C	Expans. Coeff. 11.7 ppm/°C
Trigger	Trigger Type	Auto	Encoder	Manual
	Target Window	0.5 mm	0.5 mm	0.5 mm
	Trigger Dwell	1 sec	[Not visible]	[Not visible]
Encoder	Encoder	AQB	AQB	[Not visible]
	Encoder Resolution	1 um	1 um	[Not visible]
☒ Show Hint [OK] [Cancel]				

E1733A Automation Interface Operation

Left Column Item	'CI_MEASETUP_* Constant		Right Column Item	'CI_MEASETUP_* Constant
Axis	AXIS_ITEMTEXT		Measurement Units	UNITSYSSEL_CHOOSING
Graph: Auto	AUTOTITLE_CHOOSING			
Mode	MODE_CHOOSING		Position Units	POSUNIT_CHOOSING
Cycles	CYCLES_ITEMTEXT		Error Units	ERRUNIT_CHOOSING
Position List	POSITIONS_ITEMTEXT		Display Resolution	RESULTRES_CHOOSING
Start	STARTPOS_ITEMTEXT		Averaging	AVERAGE_ITEMTEXT
End	ENDPOS_ITEMTEXT		Calibration	OCFACTOR_ITEMTEXT
Interval	INTERVAL_ITEMTEXT		Preset	PRESET_ITEMTEXT
Points	POINTS_ITEMTEXT		Dead Path	DEADPATH_ITEMTEXT
User Defined	USERDEF_CHOOSING		End Point Time Out	EPTIMEOUT_ITEMTEXT
Trigger	TRIGTYPE_CHOOSING		Keyboard Position	ALLOWKBPOS_CHOOSING
Target Window	TRIGWND_ITEMTEXT		Expansion Coeff	EXPCOEF_ITEMTEXT
Trigger Dwell	TRIGDWELL_ITEMTEXT			
Encoder Mode	AQBMODE_CHOOSING			
Encoder Res	AQBRES_ITEMTEXT		Encoder Units	AQBRES_ITEMUNIT
Apply (button)	E1733A_CC_SETUP with E1733A_CI_APPLY			--
Show Hint (chk box)	E1733A_CI_TOPSETUP_SHOWHINT_CHOOSING			0/1 for unchecked/checked
OK (button)	E1733A_CC_EXIT with E1733A_CI_SETUP			--
Cancel (button)	No Command Available			--

E1733A Automation Interface Operation

Angular Setup:

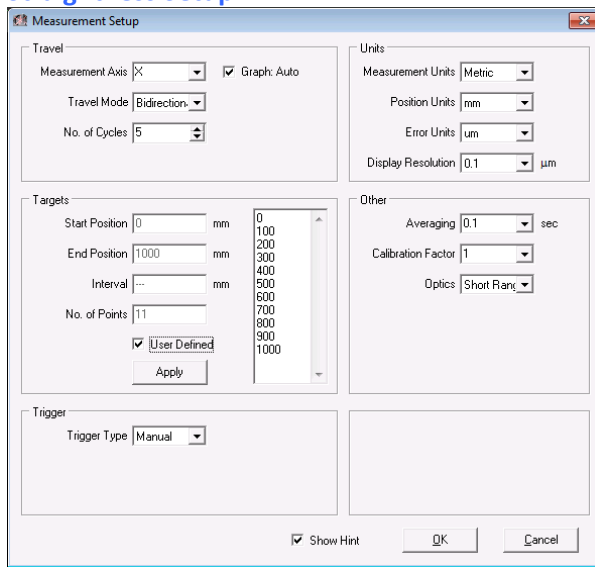


The image displays two side-by-side screenshots of the 'Measurement Setup' dialog box. Both windows have a title bar with a close button. The left window is titled 'Measurement Setup' and shows the following settings: Measurement Axis: X, Graph: Auto (checked), Travel Mode: Bidirection, No. of Cycles: 5, Measurement Units: Metric, Position Units: mm, Error Units: arcsec, Display Resolution: 0.01 arcsec. The Targets section shows Start Position: 0 mm, End Position: 1000 mm, Interval: mm, No. of Points: 11, and a checked 'User Defined' checkbox. The Trigger section shows Trigger Type: Manual. The right window is also titled 'Measurement Setup' and shows similar settings, but with Trigger Type: Encoder, Encoder: AQB, and Encoder Resolution: 1 um. Both windows have 'Show Hint', 'OK', and 'Cancel' buttons at the bottom.

Left Column Item	'CI_MEASETUP_* Constant	Right Column Item	'CI_MEASETUP_* Constant
Axis	AXIS_ITEMTEXT	Measurement Units	UNITSYSEL_CHOOSING
Graph: Auto	AUTOTITLE_CHOOSING		
Mode	MODE_CHOOSING	Position Units	POSUNIT_CHOOSING
Cycles	CYCLES_ITEMTEXT	Error Units	ERRUNIT_CHOOSING
Position List	POSITIONS_ITEMTEXT	Display Resolution	RESULTRES_CHOOSING
Start	STARTPOS_ITEMTEXT	Averaging	AVERAGE_ITEMTEXT
End	ENDPOS_ITEMTEXT	Calibration	OCFACTOR_ITEMTEXT
Interval	INTERVAL_ITEMTEXT		
Points	POINTS_ITEMTEXT		
User Defined	USERDEF_CHOOSING		
Trigger	TRIGTYPE_CHOOSING		
Encoder Mode	AQBMODE_CHOOSING		
Encoder Res	AQBRES_ITEMTEXT	Encoder Units	AQBRES_ITEMUNIT
Apply (button)	E1733A_CC_SETUP with E1733A_CI_APPLY		--
Show Hint (chk box)	E1733A_CI_TOPSETUP_SHOWHINT_CHOOSING		0/1 for unchecked/checked
OK (button)	E1733A_CC_EXIT with E1733A_CI_SETUP		--
Cancel (button)	No Command Available		--

E1733A Automation Interface Operation

Straightness Setup:



Measurement Setup

Travel
 Measurement Axis: X
☒ Graph: Auto
 Travel Mode: Bidirection
 No. of Cycles: 5

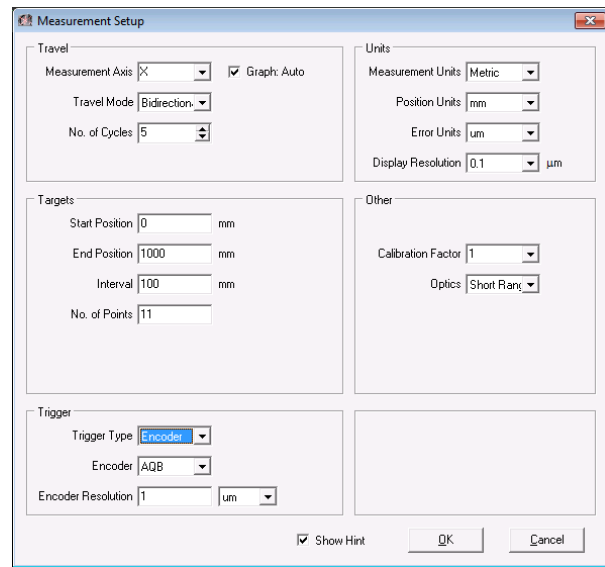
Units
 Measurement Units: Metric
 Position Units: mm
 Error Units: μm
 Display Resolution: 0.1 μm

Targets
 Start Position: 0 mm
 End Position: 1000 mm
 Interval: --- mm
 No. of Points: 11
☒ User Defined
 Apply

Other
 Averaging: 0.1 sec
 Calibration Factor: 1
 Optics: Short Range

Trigger
 Trigger Type: Manual

☒ Show Hint OK Cancel



Measurement Setup

Travel
 Measurement Axis: X
☒ Graph: Auto
 Travel Mode: Bidirection
 No. of Cycles: 5

Units
 Measurement Units: Metric
 Position Units: mm
 Error Units: μm
 Display Resolution: 0.1 μm

Targets
 Start Position: 0 mm
 End Position: 1000 mm
 Interval: 100 mm
 No. of Points: 11

Other
 Calibration Factor: 1
 Optics: Short Range

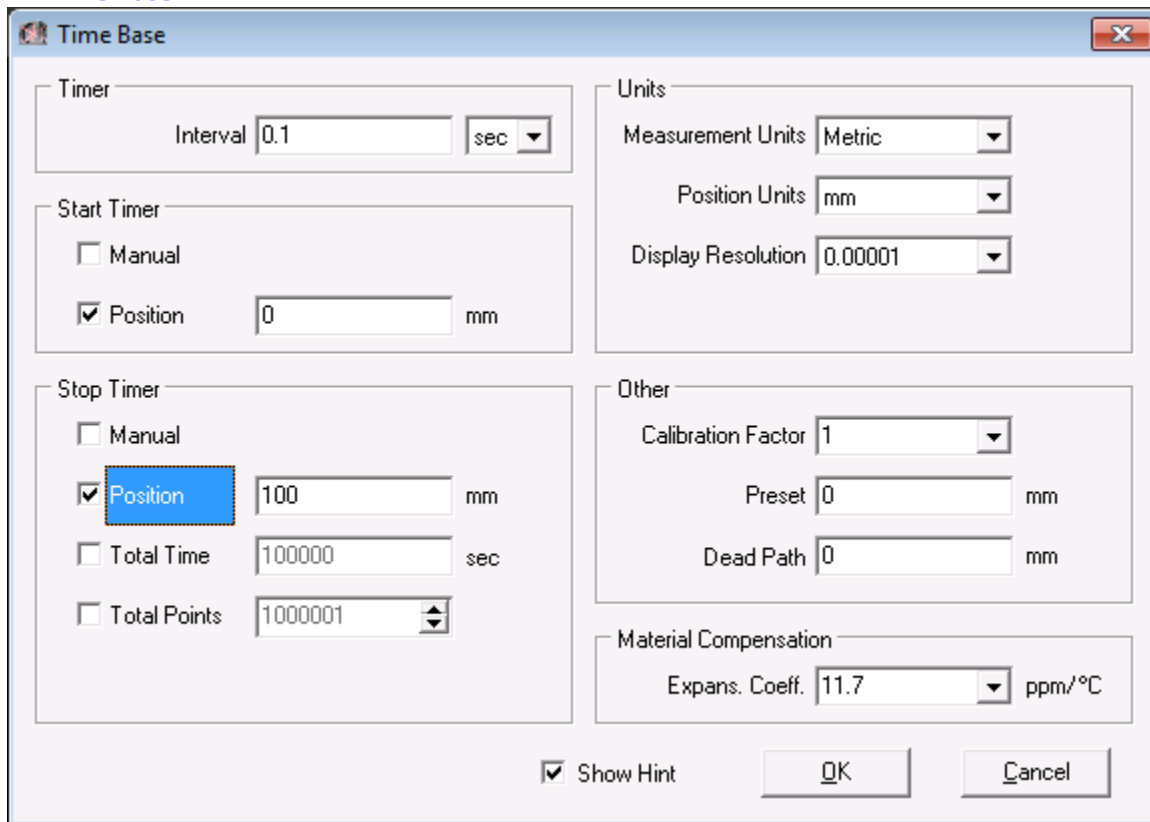
Trigger
 Trigger Type: Encoder
 Encoder: AQB
 Encoder Resolution: 1 μm

☒ Show Hint OK Cancel

Left Column Item	'CI_MEASETUP_* Constant	Right Column Item	'CI_MEASETUP_* Constant
Axis	AXIS_ITEMTEXT	Measurement Units	UNITSYSEL_CHOOSING
Graph: Auto	AUTOTITLE_CHOOSING		
Mode	MODE_CHOOSING	Position Units	POSUNIT_CHOOSING
Cycles	CYCLES_ITEMTEXT	Error Units	ERRUNIT_CHOOSING
Position List	POSITIONS_ITEMTEXT	Display Resolution	RESULTRES_CHOOSING
Start	STARTPOS_ITEMTEXT	Averaging	AVERAGE_ITEMTEXT
End	ENDPOS_ITEMTEXT	Calibration	OCFACTOR_ITEMTEXT
Interval	INTERVAL_ITEMTEXT	Optics Range	RANGETYPE_CHOOSING
Points	POINTS_ITEMTEXT		
User Defined	USERDEF_CHOOSING		
Trigger	TRIGTYPE_CHOOSING		
Encoder Mode	AQBMODE_CHOOSING		
Encoder Res	AQBRES_ITEMTEXT	Encoder Units	AQBRES_ITEMUNIT
Apply (button)	E1733A_CC_SETUP with E1733A_CI_APPLY		--
Show Hint (chk box)	E1733A_CI_TOPSETUP_SHOWHINT_CHOOSING		0/1 for unchecked/checked
OK (button)	E1733A_CC_EXIT with E1733A_CI_SETUP		--
Cancel (button)	No Command Available		--

E1733A Automation Interface Operation

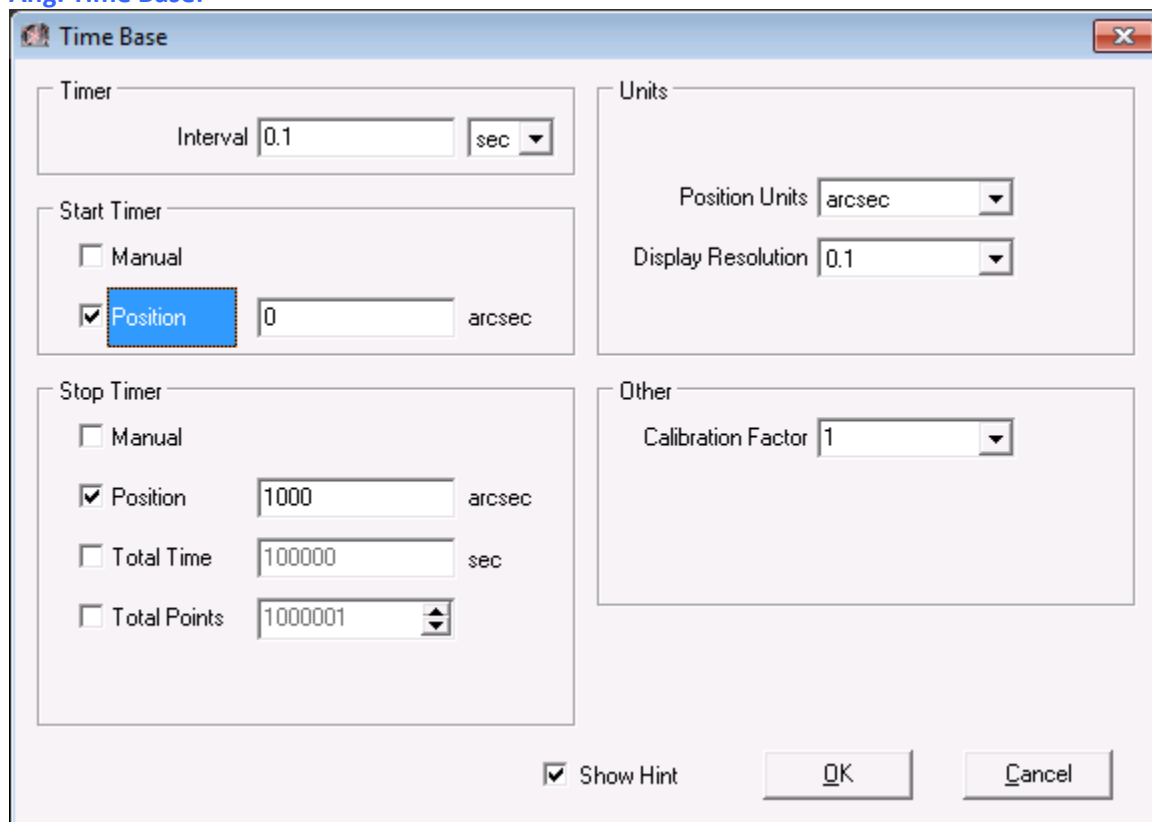
Lin Time Base:



Left Column Item	'CI_TIMSETUP_* Constant	Right Column Item	'CI_MEASETUP_* Constant
Interval	TBINTERVAL_ITEMTEXT	Measurement Units	UNITSYSSEL_CHOOSING
Interval Units	TBINTERVAL_ITEMUNIT	Position Units	POSUNIT_CHOOSING
Start Type	STARTTYPE_CHOOSING	Display Resolution	RESULTRES_CHOOSING
Start Position	TBSTARTPOS_ITEMTEXT		
Stop Type	STOPTYPE_CHOOSING	Calibration	OCFACTOR_ITEMTEXT
Stop Position	TBSTOPPOS_ITEMTEXT	Preset	PRESET_ITEMTEXT
Total Time	TOTALTIME_ITEMTEXT	Dead Path	DEADPATH_ITEMTEXT
Total Points	POINTNUM_ITEMTEXT	Expansion Coeff.	EXPCOEF_ITEMTEXT
Show Hint (chk box)	E1733A_CI_TOPSETUP_SHOWHINT_CHOOSING		0/1 for unchecked/checked
OK (button)	E1733A_CC_EXIT with E1733A_CI_SETUP		--
Cancel (button)	No Command Available		--

E1733A Automation Interface Operation

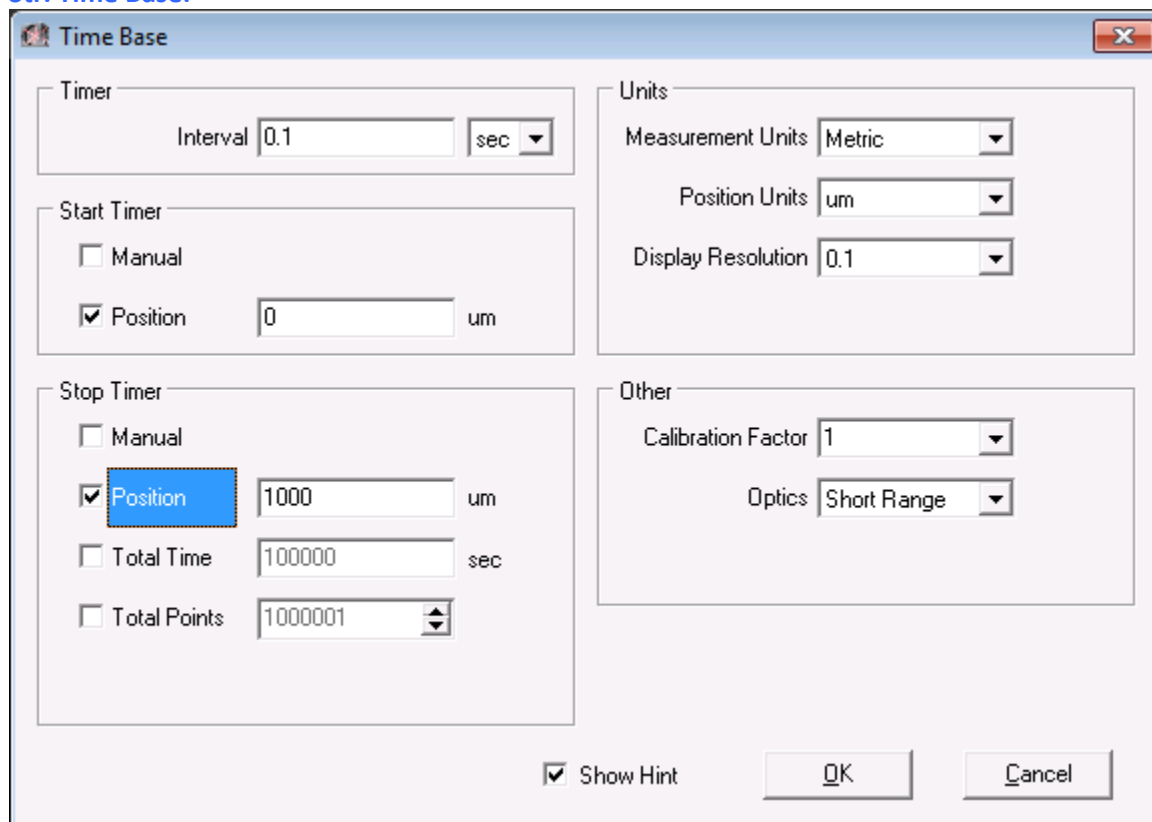
Ang. Time Base:



Left Column Item	'CI_TIMSETUP_* Constant		Right Column Item	'CI_MEASETUP_* Constant
Interval	TBINTERVAL_ITEMTEXT		Position Units	POSUNIT_CHOOSING
Interval Units	TBINTERVAL_ITEMUNIT		Display Resolution	RESULTRES_CHOOSING
Start Type	STARTTYPE_CHOOSING			
Start Position	TBSTARTPOS_ITEMTEXT			
Stop Type	STOPTYPE_CHOOSING		Calibration	OCFACTOR_ITEMTEXT
Stop Position	TBSTOPPOS_ITEMTEXT			
Total Time	TOTALTIME_ITEMTEXT			
Total Points	POINTNUM_ITEMTEXT			
Show Hint (chk box)	E1733A_CI_TOPSETUP_SHOWHINT_CHOOSING			0/1 for unchecked/checked
OK (button)	E1733A_CC_EXIT with E1733A_CI_SETUP			--
Cancel (button)	No Command Available			--

E1733A Automation Interface Operation

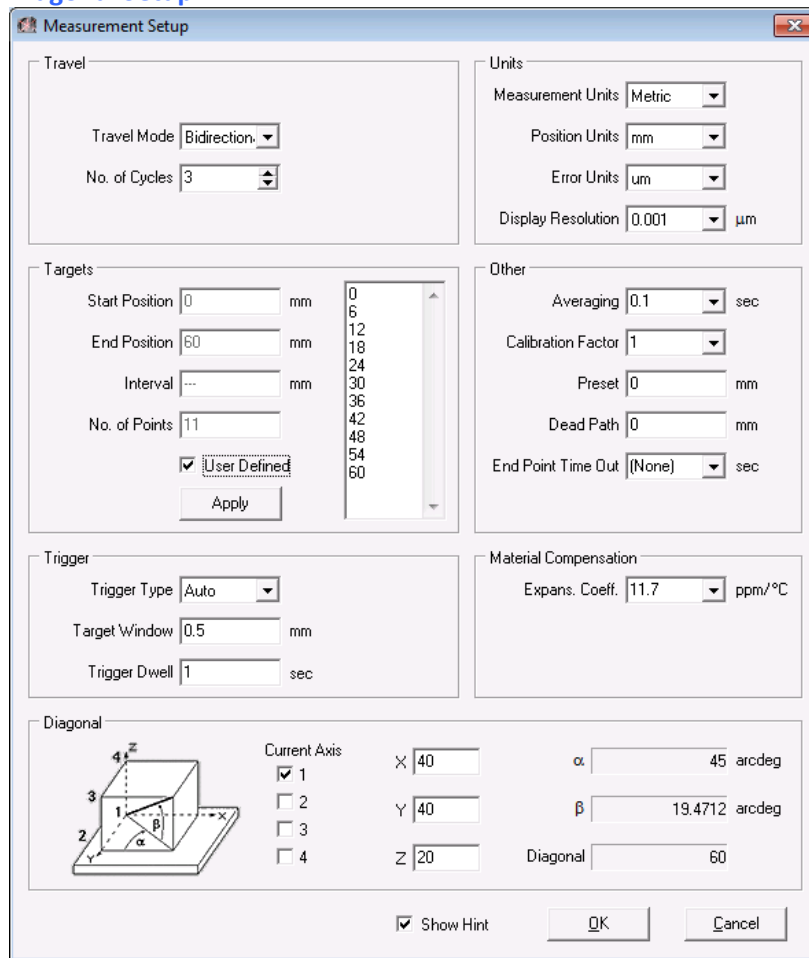
Str. Time Base:



Left Column Item	'CI_TIMSETUP_* Constant	Right Column Item	'CI_MEASETUP_* Constant
Interval	TBINTERVAL_ITEMTEXT	Measurement Units	UNITSYSSEL_CHOOSING
Interval Units	TBINTERVAL_ITEMUNIT	Position Units	POSUNIT_CHOOSING
Start Type	STARTTYPE_CHOOSING	Display Resolution	RESULTRES_CHOOSING
Start Position	TBSTARTPOS_ITEMTEXT		
Stop Type	STOPTYPE_CHOOSING	Calibration	OCFACTOR_ITEMTEXT
Stop Position	TBSTOPPOS_ITEMTEXT	Optics Range	RANGETYPE_CHOOSING
Total Time	TOTALTIME_ITEMTEXT		
Total Points	POINTNUM_ITEMTEXT		
Show Hint (chk box)	E1733A_CI_TOPSETUP_SHOWHINT_CHOOSING		0/1 for unchecked/checked
OK (button)	E1733A_CC_EXIT with E1733A_CI_SETUP		--
Cancel (button)	No Command Available		--

E1733A Automation Interface Operation

Diagonal Setup:



Measurement Setup

Travel
Travel Mode: Bidirection
No. of Cycles: 3

Units
Measurement Units: Metric
Position Units: mm
Error Units: um
Display Resolution: 0.001 um

Targets
Start Position: 0 mm
End Position: 60 mm
Interval: --- mm
No. of Points: 11
☒ User Defined
Apply

Other
Averaging: 0.1 sec
Calibration Factor: 1
Preset: 0 mm
Dead Path: 0 mm
End Point Time Out: (None) sec

Trigger
Trigger Type: Auto
Target Window: 0.5 mm
Trigger Dwell: 1 sec

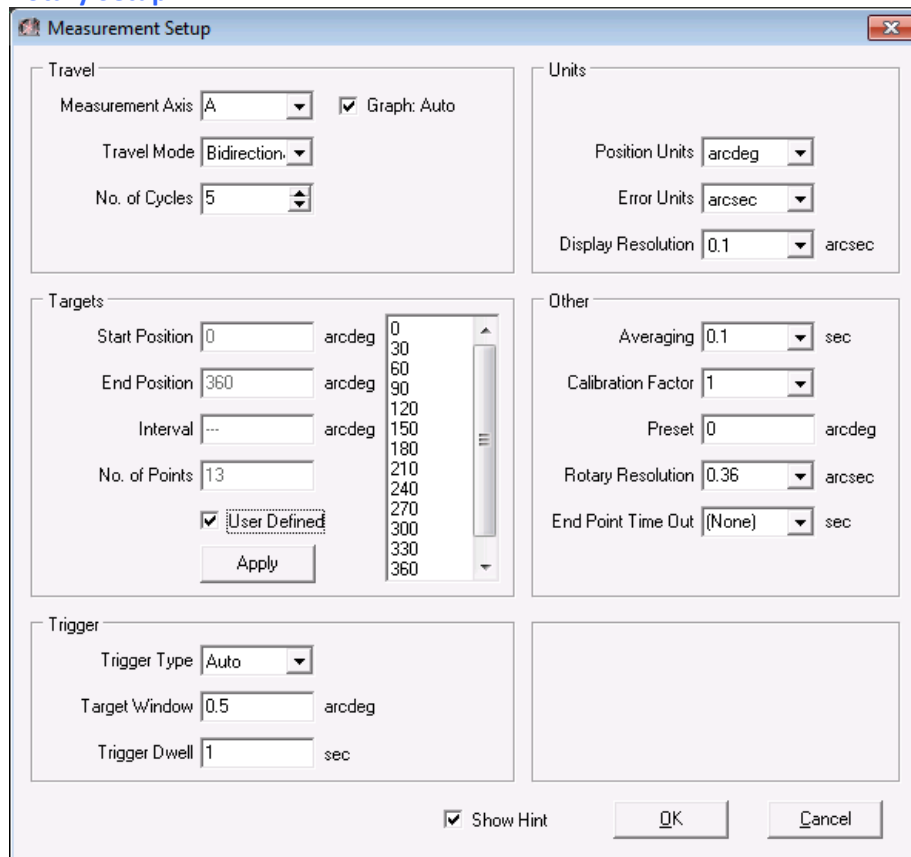
Material Compensation
Expans. Coeff: 11.7 ppm/°C

Diagonal
Current Axis:
☒ 1
☐ 2
☐ 3
☐ 4
X: 40
Y: 40
Z: 20
Diagonal: 60
alpha: 45 arcdeg
beta: 19.4712 arcdeg
☒ Show Hint
OK Cancel

Left Column Item	'CI_MEASETUP_* Constant	Right Column Item	'CI_MEASETUP_* Constant
Mode	MODE_CHOOSING	Measurement Units	UNITSYSEL_CHOOSING
Cycles	CYCLES_ITEMTEXT	Position Units	POSUNIT_CHOOSING
		Error Units	ERRUNIT_CHOOSING
Position List	POSITIONS_ITEMTEXT	Display Resolution	RESULTRES_CHOOSING
Start	STARTPOS_ITEMTEXT	Averaging	AVERAGE_ITEMTEXT
End	ENDPOS_ITEMTEXT	Calibration	OCFACTOR_ITEMTEXT
Interval	INTERVAL_ITEMTEXT	Preset	PRESET_ITEMTEXT
Points	POINTS_ITEMTEXT	Dead Path	DEADPATH_ITEMTEXT
User Defined	USERDEF_CHOOSING	End Point Time Out	EPTIMEOUT_ITEMTEXT
Trigger	TRIGTYPE_CHOOSING	Expansion Coeff	EXPCOEF_ITEMTEXT
Target Window	TRIGWND_ITEMTEXT	X Length	DIASCALEX_ITEMTEXT
Trigger Dwell	TRIGDWELL_ITEMTEXT	Y Length	DIASCALEY_ITEMTEXT
Current Axis	MEAXISSEL_CHOOSING	Z Length	DIASCALEZ_ITEMTEXT
Apply (button)	E1733A_CC_SETUP with E1733A_CI_APPLY		--
Show Hint (chk box)	E1733A_CI_TOPSETUP_SHOWHINT_CHOOSING		0/1 for unchecked/checked
OK (button)	E1733A_CC_EXIT with E1733A_CI_SETUP		--
Cancel (button)	No Command Available		--

E1733A Automation Interface Operation

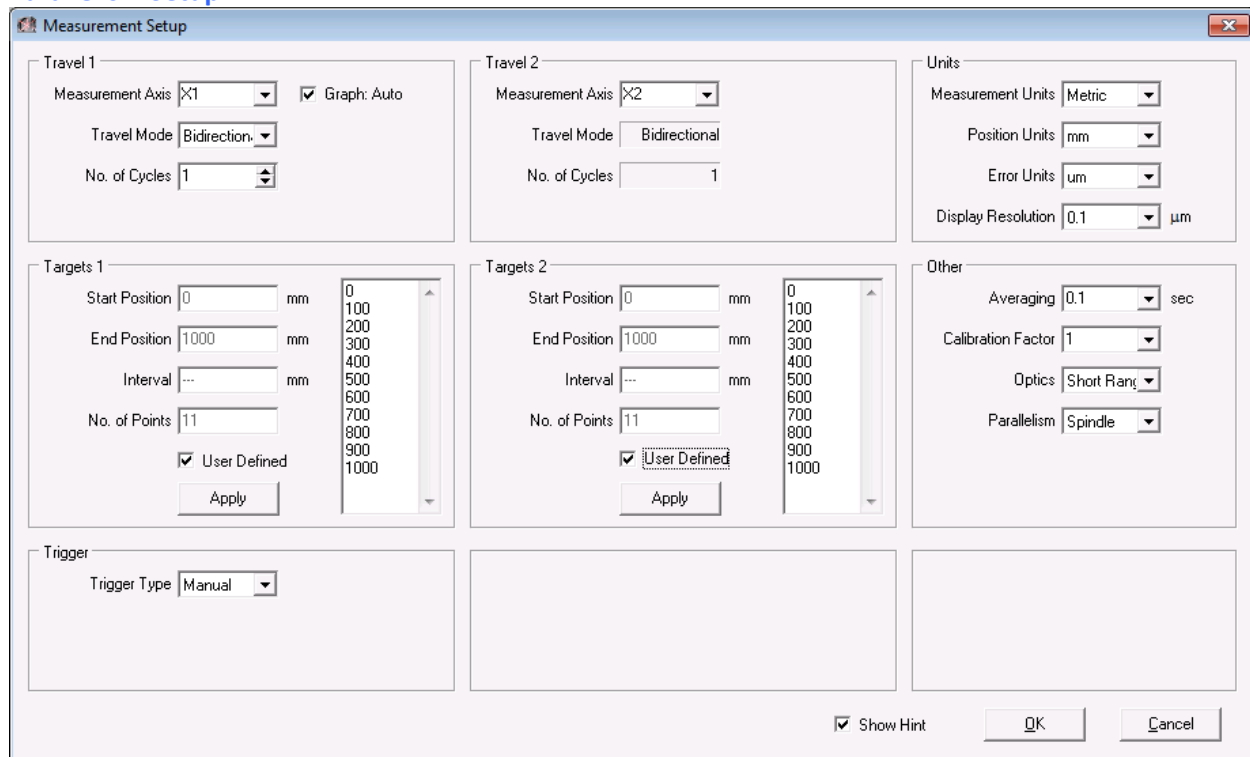
Rotary Setup:



Left Column Item	'CI_MEASETUP_* Constant	Right Column Item	'CI_MEASETUP_* Constant
Axis	AXIS_ITEMTEXT	Position Units	POSUNIT_CHOOSING
Graph: Auto	AUTOTITLE_CHOOSING		
Mode	MODE_CHOOSING	Error Units	ERRUNIT_CHOOSING
Cycles	CYCLES_ITEMTEXT	Display Resolution	RESULTRES_CHOOSING
Position List	POSITIONS_ITEMTEXT		
Start	STARTPOS_ITEMTEXT	Averaging	AVERAGE_ITEMTEXT
End	ENDPOS_ITEMTEXT	Calibration	OCFACTOR_ITEMTEXT
Interval	INTERVAL_ITEMTEXT	Preset	PRESET_ITEMTEXT
Points	POINTS_ITEMTEXT	Rotary Resolution	ROTAQBRES_ITEMTEXT
User Defined	USERDEF_CHOOSING	End Point Time Out	EPTIMEOUT_ITEMTEXT
Trigger	TRIGTYPE_CHOOSING		
Target Window	TRIGWND_ITEMTEXT		
Trigger Dwell	TRIGDWELL_ITEMTEXT		
Apply (button)	E1733A_CC_SETUP with E1733A_CI_APPLY		--
Show Hint (chk box)	E1733A_CI_TOPSETUP_SHOWHINT_CHOOSING		0/1 for unchecked/checked
OK (button)	E1733A_CC_EXIT with E1733A_CI_SETUP		--
Cancel (button)	No Command Available		--

E1733A Automation Interface Operation

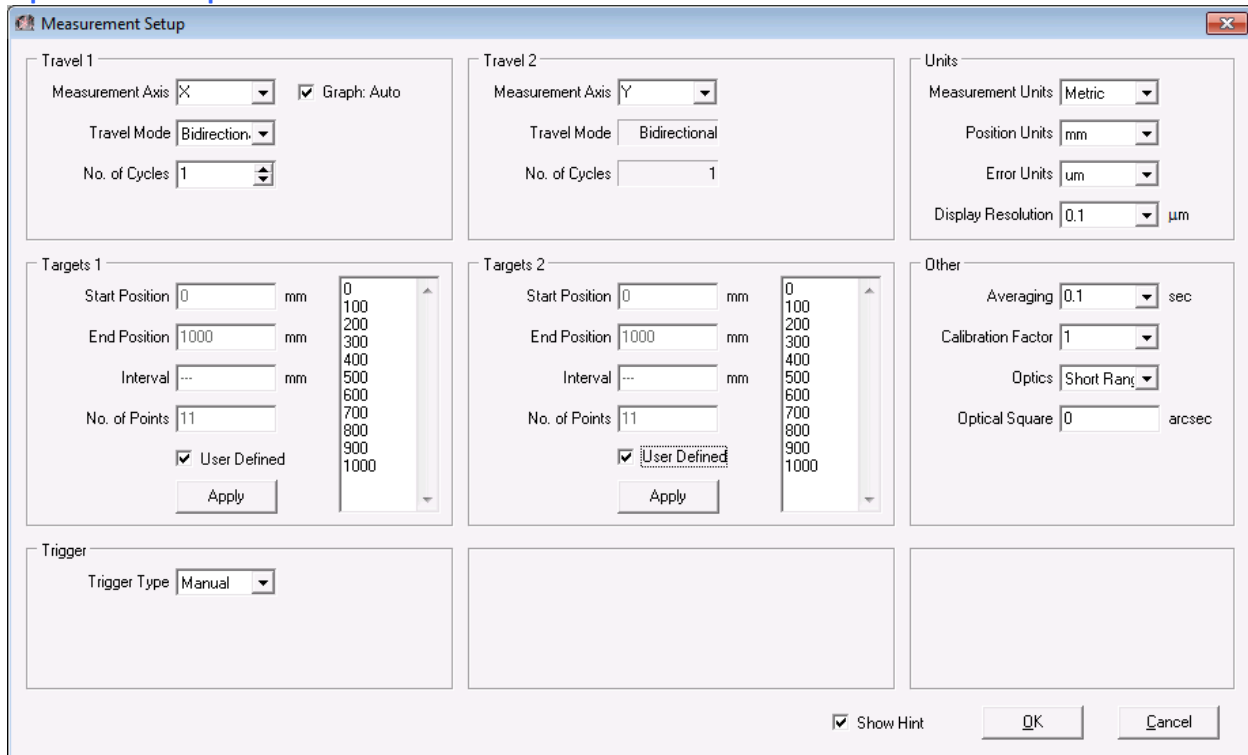
Parallelism Setup:



Left Column Item	'CI_MEASETUP_* Constant (use char in [] for 2 nd axis)	Right Column Item	'CI_MEASETUP_* Constant
Axis	AXIS[2]_ITEMTEXT	Measurement Units	UNITSYSEL_CHOOSING
Graph: Auto	AUTOTITLE_CHOOSING		
Mode	MODE_CHOOSING	Position Units	POSUNIT_CHOOSING
Cycles	CYCLES_ITEMTEXT	Error Units	ERRUNIT_CHOOSING
Position List	POSITIONS[2]_ITEMTEXT	Display Resolution	RESULTRES_CHOOSING
Start	STARTPOS[2]_ITEMTEXT	Averaging	AVERAGE_ITEMTEXT
End	ENDPOS[2]_ITEMTEXT	Calibration	OCFACTOR_ITEMTEXT
Interval	INTERVAL[2]_ITEMTEXT	Optics Range	RANGETYPE_CHOOSING
Points	POINTS[2]_ITEMTEXT	Parallelism Mode	PARATYPE_CHOOSING
User Defined	USERDEF[2]_CHOOSING		
Trigger	TRIGTYPE_CHOOSING		
Apply (button)	E1733A_CC_SETUP with E1733A_CI_APPLY[2]		
Show Hint (chk box)	E1733A_CI_TOPSETUP_SHOWHINT_CHOOSING		0/1 for unchecked/checked
OK (button)	E1733A_CC_EXIT with E1733A_CI_SETUP		--
Cancel (button)	No Command Available		--

E1733A Automation Interface Operation

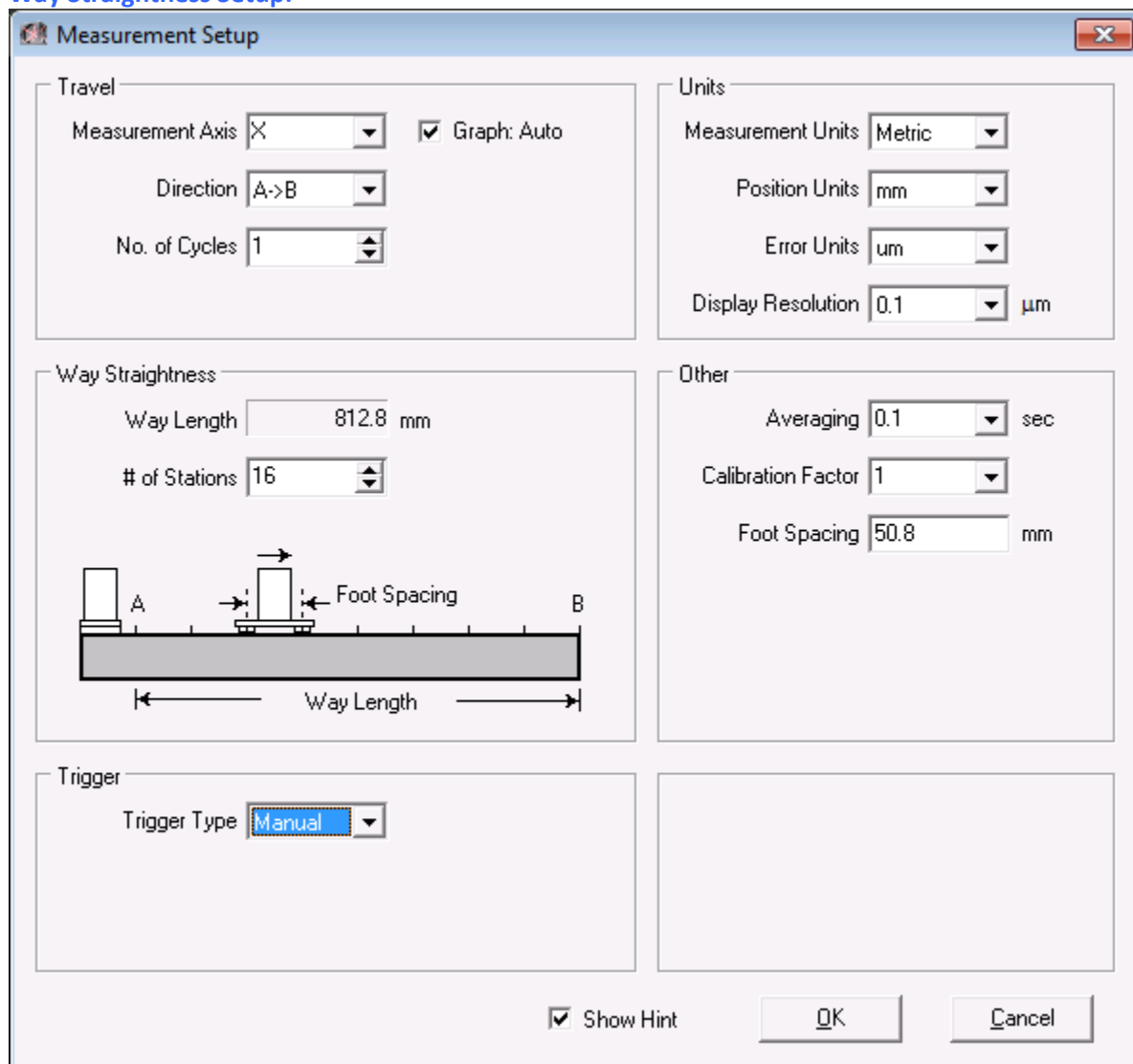
Squareness Setup:



Left Column Item	'CI_MEASETUP_* Constant (use char in [] for 2 nd axis)	Right Column Item	'CI_MEASETUP_* Constant
Axis	AXIS[2]_ITEMTEXT	Measurement Units	UNITSYSSEL_CHOOSING
Graph: Auto	AUTOTITLE_CHOOSING		
Mode	MODE_CHOOSING	Position Units	POSUNIT_CHOOSING
Cycles	CYCLES_ITEMTEXT	Error Units	ERRUNIT_CHOOSING
Position List	POSITIONS[2]_ITEMTEXT	Display Resolution	RESULTRES_CHOOSING
Start	STARTPOS[2]_ITEMTEXT	Averaging	AVERAGE_ITEMTEXT
End	ENDPOS[2]_ITEMTEXT	Calibration	OCFACTOR_ITEMTEXT
Interval	INTERVAL[2]_ITEMTEXT	Optics Range	RANGETYPE_CHOOSING
Points	POINTS[2]_ITEMTEXT	Optical Square	SQUAREERR_ITEMTEXT
User Defined	USERDEF[2]_CHOOSING		
Trigger	TRIGTYPE_CHOOSING		
Apply (button)	E1733A_CC_SETUP with E1733A_CI_APPLY[2]		
Show Hint (chk box)	E1733A_CI_TOPSETUP_SHOWHINT_CHOOSING		0/1 for unchecked/checked
OK (button)	E1733A_CC_EXIT with E1733A_CI_SETUP		--
Cancel (button)	No Command Available		--

E1733A Automation Interface Operation

Way Straightness Setup:



Measurement Setup

Travel

Measurement Axis: ☒ Graph: Auto

Direction:

No. of Cycles:

Units

Measurement Units:

Position Units:

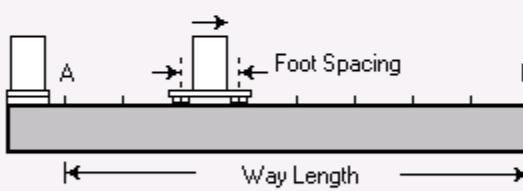
Error Units:

Display Resolution: μm

Way Straightness

Way Length: mm

of Stations:



Other

Averaging: sec

Calibration Factor:

Foot Spacing: mm

Trigger

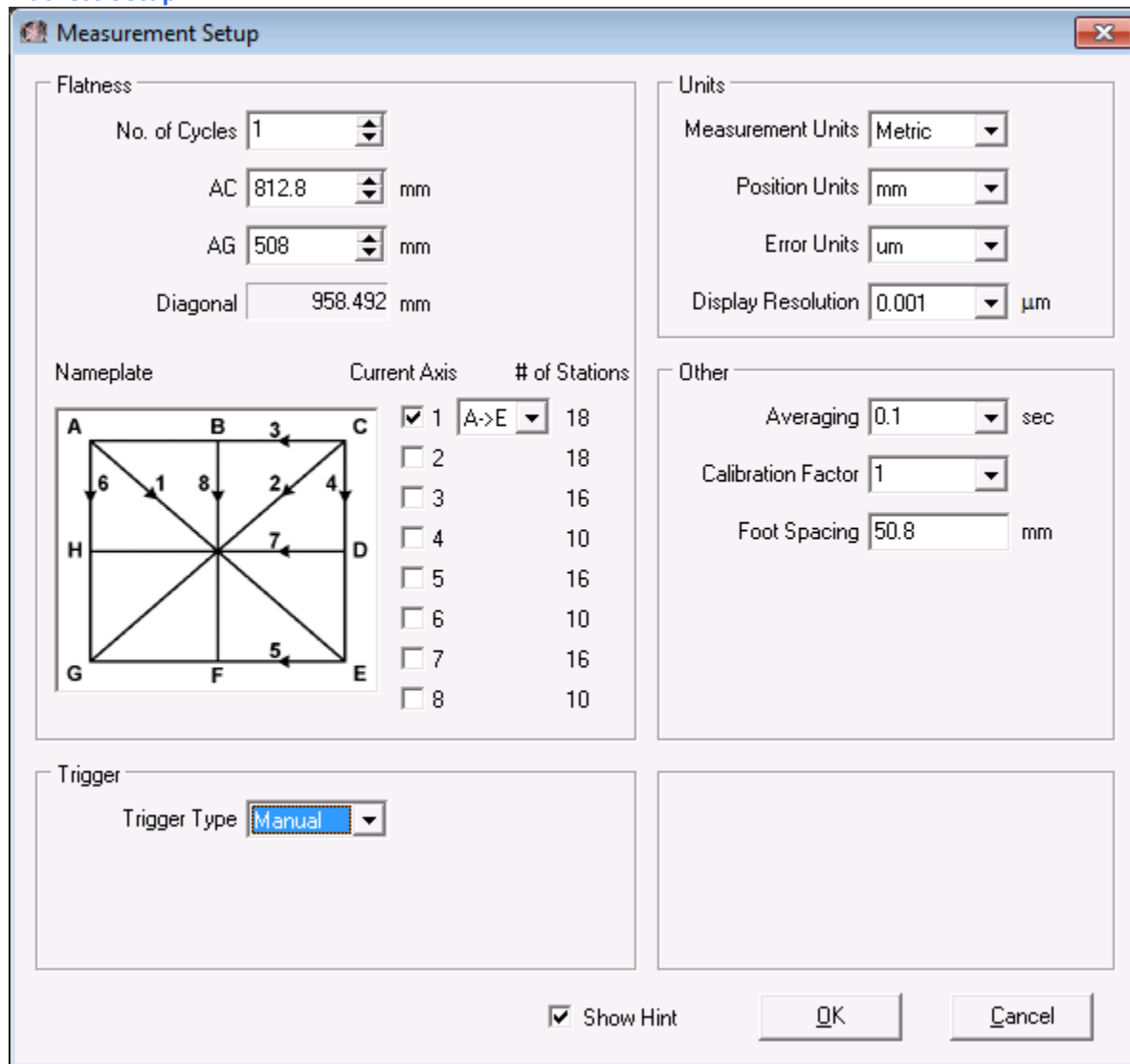
Trigger Type:

☒ Show Hint

Left Column Item	'CI_MEASETUP_* Constant	Right Column Item	'CI_MEASETUP_* Constant
Axis	AXIS_ITEMTEXT	Measurement Units	UNITSYSSEL_CHOOSING
Graph: Auto	AUTOTITLE_CHOOSING		
Direction	WAYDIR_CHOOSING	Position Units	POSUNIT_CHOOSING
Cycles	CYCLES_ITEMTEXT	Error Units	ERRUNIT_CHOOSING
		Display Resolution	RESULTRES_CHOOSING
Stations	WAYSTANUM_ITEMTEXT	Averaging	AVERAGE_ITEMTEXT
		Calibration	OCTFACTOR_ITEMTEXT
		Foot Spacing	FOOTSPACE_ITEMTEXT
Trigger	TRIGTYPE_CHOOSING		
Show Hint (chk box)	E1733A_CI_TOPSETUP_SHOWHINT_CHOOSING		0/1 for unchecked/checked
OK (button)	E1733A_CC_EXIT with E1733A_CI_SETUP		--
Cancel (button)	No Command Available		--

E1733A Automation Interface Operation

Flatness Setup:



Measurement Setup

Flatness

No. of Cycles: 1

AC: 812.8 mm

AG: 508 mm

Diagonal: 958.492 mm

Units

Measurement Units: Metric

Position Units: mm

Error Units: μ m

Display Resolution: 0.001 μ m

Nameplate

Current Axis

of Stations

Station	Current Axis	# of Stations
1	A->E	18
2		18
3		16
4		10
5		16
6		10
7		16
8		10

Other

Averaging: 0.1 sec

Calibration Factor: 1

Foot Spacing: 50.8 mm

Trigger

Trigger Type: Manual

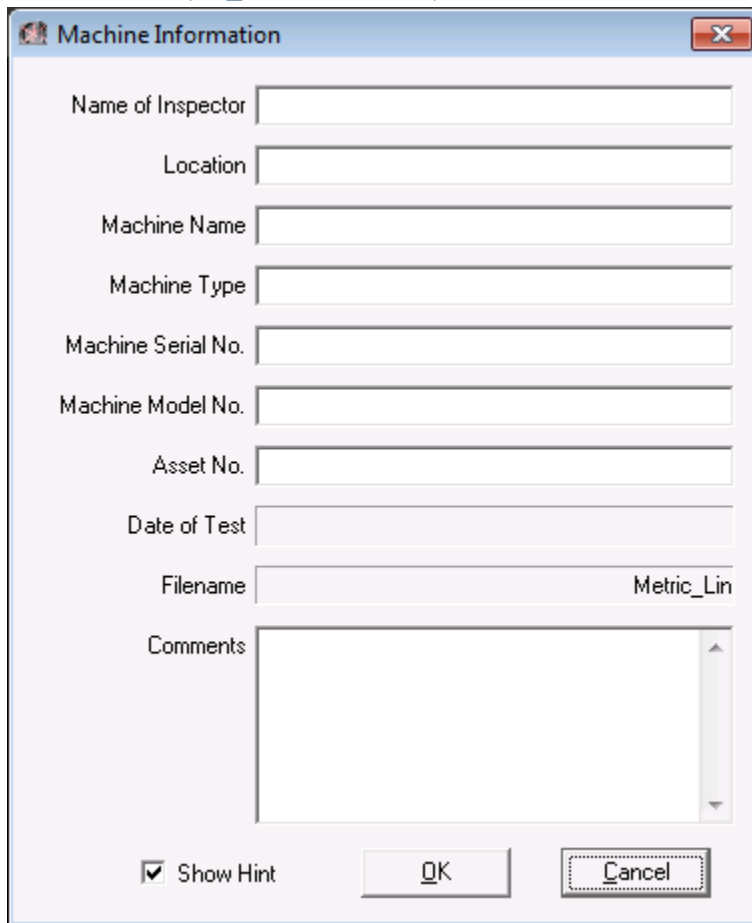
☒ Show Hint

OK Cancel

Left Column Item	'CI_MEASETUP_* Constant	Right Column Item	'CI_MEASETUP_* Constant
Cycles	CYCLES_ITEMTEXT	Measurement Units	UNITSYSSEL_CHOOSING
AC Length	FLALENAC_ITEMTEXT	Position Units	POSUNIT_CHOOSING
AG Length	FLALENAG_ITEMTEXT	Error Units	ERRUNIT_CHOOSING
		Display Resolution	RESULTRES_CHOOSING
Axis Select	MEAAXISSEL_CHOOSING	Averaging	AVERAGE_ITEMTEXT
Axis Direction	FLAAXISDIR_CHOOSING	Calibration	OCCFACTOR_ITEMTEXT
Stations A2C	FLASTAAC_ITEMTEXT	Foot Spacing	FOOTSPACE_ITEMTEXT
Stations A2G	FLASTAAG_ITEMTEXT		
Trigger	TRIGTYPE_CHOOSING		
Show Hint (chk box)	E1733A_CI_TOPSETUP_SHOWHINT_CHOOSING		0/1 for unchecked/checked
OK (button)	E1733A_CC_EXIT with E1733A_CI_SETUP		--
Cancel (button)	No Command Available		--

E1733A Automation Interface Operation

Machine Info ('CI_MACHINEINFO):



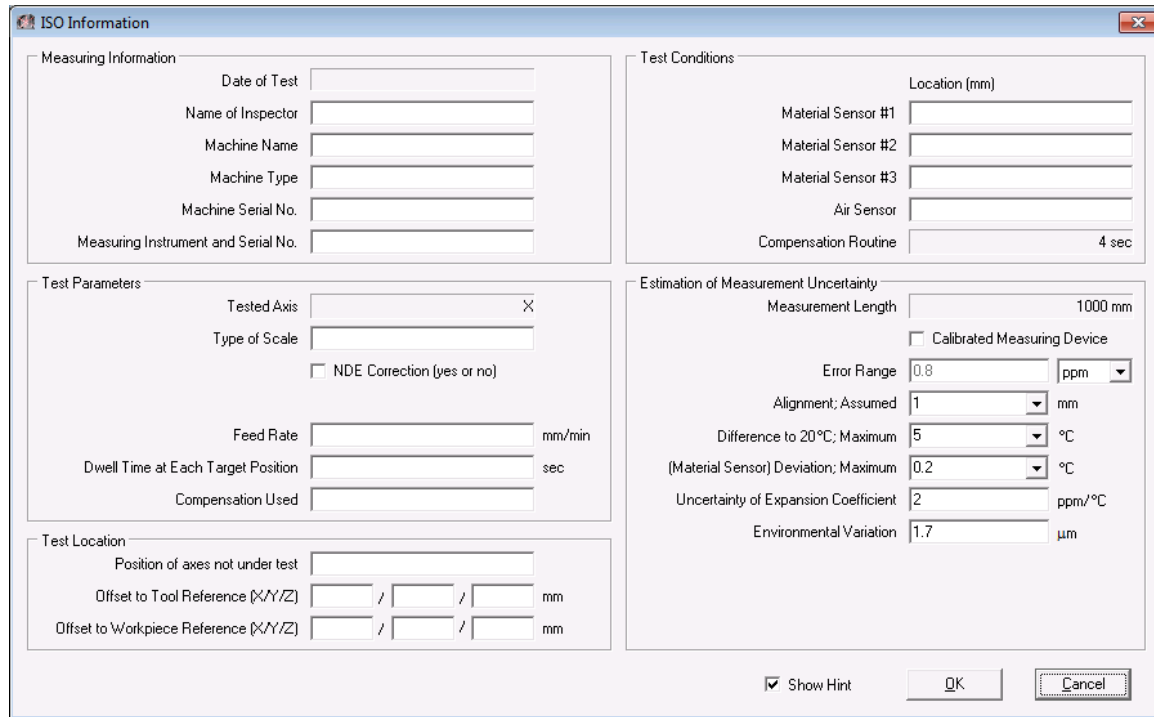
The image shows a 'Machine Information' dialog box with the following fields and controls:

- Name of Inspector: [Text Field]
- Location: [Text Field]
- Machine Name: [Text Field]
- Machine Type: [Text Field]
- Machine Serial No.: [Text Field]
- Machine Model No.: [Text Field]
- Asset No.: [Text Field]
- Date of Test: [Text Field]
- Filename: [Text Field] Metric_Lin: [Text Field]
- Comments: [Text Area]
- ☒ Show Hint
- OK button
- Cancel button

Item	'CI_INFSETUP_* Constant	What to put in shared memory
Inspector Name	OPERATOR_ITEMTEXT	Desired value as text string
Location	LOCATION_ITEMTEXT	
Machine Name	MACHNAME_ITEMTEXT	
Machine Type	MACHTYPE_ITEMTEXT	
Machine Serial #	MACHSN_ITEMTEXT	
Machine Model #	MACHMODEL_ITEMTEXT	
Asset #	ASSETNO_ITEMTEXT	
Date of Test	DATETEST_ITEMTEXT	Read Only
File Name	--	Not available
Comments	COMMENTS_ITEMTEXT	Desired lines as list of comma separated quoted strings.
Show Hint (chk box)	E1733A_CI_TOPSETUP_SHOWHINT_CHOOSING	0/1 for unchecked/checked
OK (button)	E1733A_CC_EXIT with E1733A_CI_SETUP	--
Cancel (button)	No Command Available	--

E1733A Automation Interface Operation

ISO Info ('CI_ISOINFO):



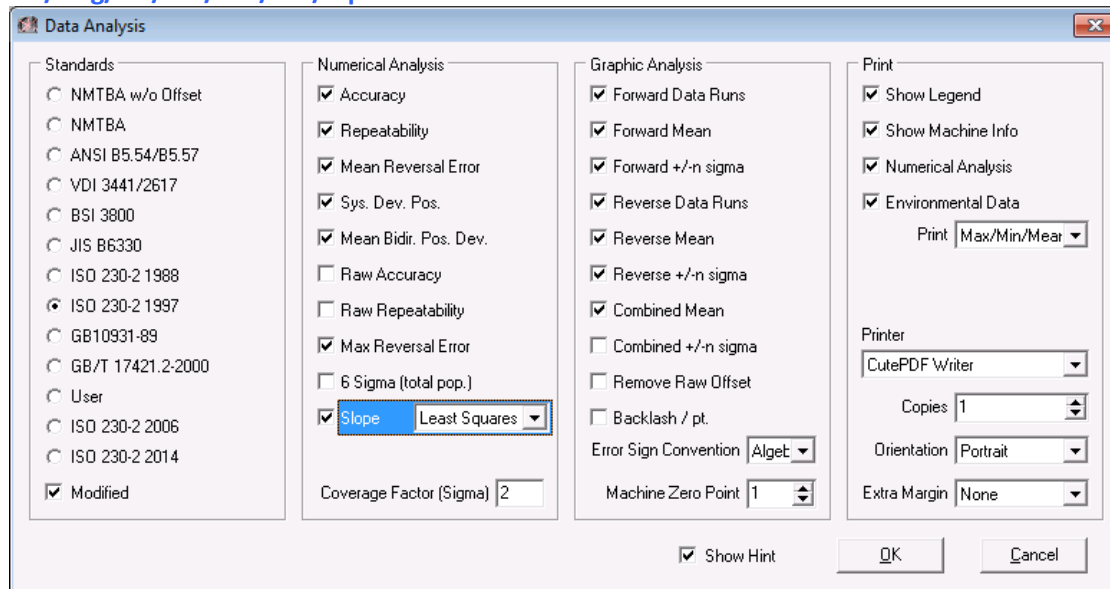
Left Column Item	'CI_ISOSETUP_* Constant	Right Column Item	'CI_ISOSETUP_* Constant
Inspector Name**	OPERATOR_ITEMTEXT	MT1 Location	MATSEN1_ITEMTEXT
Machine Name**	MACHNAME_ITEMTEXT	MT2 Location	MATSEN2_ITEMTEXT
Machine Type**	MACHTYPE_ITEMTEXT	MT3 Location	MATSEN3_ITEMTEXT
Machine Serial**	MACHSN_ITEMTEXT	Air Sensor Location	ATSEN_ITEMTEXT
Meas. Inst and SN	MEAINST_ITEMTEXT		
Tested Axis	E1733A_CI_MEASETUP_AXIS_ITEMTEXT		
Type of Scale	TYPESCALE_ITEMTEXT	Calibrated Device	CALMEADEV_CHOOSING
NDE Correction	NDECORR_CHOOSING	Error Range	ERRORRANGE_ITEMTEXT
CTE of Scale	COEFSCALE_ITEMTEXT	Error Range Units	ERRORRANGE_ITEMUNIT
Feed Rate	FEEDRATE_ITEMTEXT	Alignment	ALIGNERR_ITEMTEXT
Dwell Time	DWELLTIME_ITEMTEXT	Diff to 20C	DIFF20CMAX_ITEMTEXT
Compensation	COMPUSED_ITEMTEXT	Mat Sense Deviation	MATTEMPDEV_ITEMTEXT
Other axes position	POSAXES_ITEMTEXT	TCE uncertainty	UNCEXPCOEF_ITEMTEXT
Offset to tool Ref X	TOOLX_ITEMTEXT	Env. Variation	ENVVARERR_ITEMTEXT
Offset to tool Ref Y	TOOLY_ITEMTEXT		
Offset to tool Ref Z	TOOLZ_ITEMTEXT		
Offset to work Ref X	PIECEX_ITEMTEXT		
Offset to work Ref Y	PIECEY_ITEMTEXT		
Offset to work Ref Z	PIECEZ_ITEMTEXT		
Show Hint (chk box)	E1733A_CI_TOPSETUP_SHOWHINT_CHOOSING		0/1 for unchecked/checked
OK (button)	E1733A_CC_EXIT with E1733A_CI_SETUP		--
Cancel (button)	No Command Available		--

** Within 'CI_INFSETUP_* group instead of 'CI_ISOSETUP_* group.

E1733A Automation Interface Operation

Data Analysis Setup (CI_ANALYSIS):

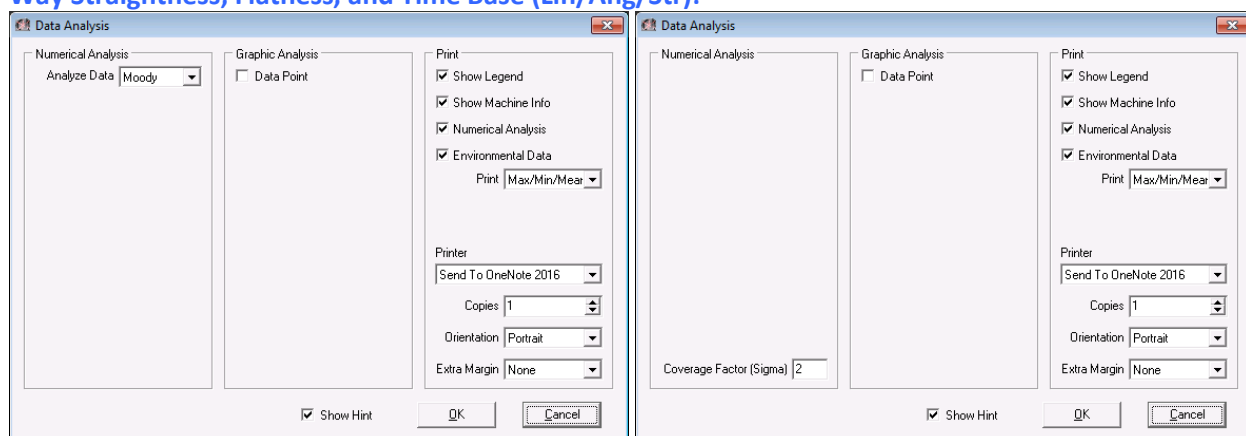
Lin/Ang/Str/Dia/Rot/Par/Squ:



Item	'CI_ANALYSIS_* Constant	What to put in shared memory
Standard Selection	STANDARD_CHOOSING	Use a # as string: 0 for 1 st thru 12 for last
Modified Checkbox	ANAMOD_CHOOSING	Use 0 to uncheck (cannot directly check, auto checked when other items changed)
Numerical Analysis Checks	ANANUM_COMPRISE	String of 0/1 chars, left most is top checkbox, missing are seen as 0, use 1 for check, 0 for uncheck
Graphic Analysis Checks	ANAGRAPH_COMPRISE	
Print Show Checkboxes	PRNSHOWSEL_COMPRISE	
Slope Type Selection	ANASLOPE_CHOOSING	Use 0 for Least Squares and 1 for End Points
Sigma Factor	COMPZERO_ITEMTEXT	Desired value as text string
Error Sign Convention	E1733A_CI_CMPSETUP_COMPABS_CHOOSING	
Machine Zero Point	ANASIGMA_ITEMTEXT	
Copes	PRNCOPIES_ITEMTEXT	
Print Data Select	PRNENVSEL_CHOOSING	Use 0 for Max/Min/Mean and 1 for Start/End
Printer	PRNNAME_CHOOSING	Use 0 ... N: Index of installed printers.
Orientation	PRNORIENT_CHOOSING	Use 0 / 1 for Portrait / Landscape
Extra Margin	PRNMARGIN_CHOOSING	Use 0 ... 4 for None, Lft, Top, Rgt, Bot
Show Hint (chk box)	E1733A_CI_TOPSETUP_SHOWHINT_CHOOSING	Use 0/1 for unchecked/checked
OK (button)	E1733A_CC_EXIT with E1733A_CI_SETUP	--
Cancel (button)	No Command Available	--

E1733A Automation Interface Operation

Way Straightness, Flatness, and Time Base (Lin/Ang/Str):



The image shows two side-by-side screenshots of the 'Data Analysis' dialog box. Both windows have a title bar with a red 'X' icon. The left window has the 'Analyze Data' dropdown menu open, showing 'Moody' selected. The right window has the 'Coverage Factor (Sigma)' dropdown menu open, showing '2' selected. Both windows have a 'Print' section with checkboxes for 'Show Legend', 'Show Machine Info', 'Numerical Analysis', and 'Environmental Data'. The 'Printer' section has a dropdown for 'Send To OneNote 2016', a 'Copies' spinner set to '1', an 'Orientation' dropdown set to 'Portrait', and an 'Extra Margin' dropdown set to 'None'. Both windows have 'Show Hint', 'OK', and 'Cancel' buttons at the bottom.

Item	'CI_ ANASETUP_* Constant	What to put in shared memory
Analyze Data	FLAMITHOD_CHOOSING	Use 0 for Moody
Data Point Checkbox	TBDATAPNT_CHOOSING	Use 0 / 1 for Unchecked / Checked
Print Show Checkboxes	PRNSHOWSEL_COMPRISE	String of 0/1 chars, left most is top checkbox, missing are seen as 0, use 1 for check, 0 for uncheck
Sigma Factor	COMPZERO_ITEMTEXT	Desired value as text string
Copes	PRNCOPIES_ITEMTEXT	
Print Data Select	PRNENVSEL_CHOOSING	Use 0 for Max/Min/Mean and 1 for Start/End
Printer	PRNNAME_CHOOSING	Use 0 ... N: Index of installed printers.
Orientation	PRNORIENT_CHOOSING	Use 0 / 1 for Portrait / Landscape
Extra Margin	PRNMARGIN_CHOOSING	Use 0 ... 4 for None, Lft, Top, Rgt, Bot
Show Hint (chk box)	E1733A_CI_TOPSETUP_SHOWHINT_CHOOSING	Use 0/1 for unchecked/checked
OK (button)	E1733A_CC_EXIT with E1733A_CI_SETUP	--
Cancel (button)	No Command Available	--

E1733A Automation Interface Operation

Data Analysis Results ('CC_ANALYSIS):

Send the 'CC_ANALYSIS command with the indicated 'CI_* constants and sometimes required NUM values to read the desired analysis result value by interpreting the returned long value as a float value.

Linear and VDI:

Data Analysis	
Standards: ISO 230-2 1997 (f	
Accuracy:	
A = 9.6 μm	
A $\uparrow$ = 6.1 μm	
A $\downarrow$ = 5.8 μm	
Sys. Dev. Pos.:	
E = 7.7 μm	
E $\uparrow$ = 3.9 μm	
E $\downarrow$ = 4.0 μm	
Mean Bidir. Pos. Dev.:	
M = 4.0 μm	
Repeatability:	
R = 6.6 μm	
R $\uparrow$ = 3.0 μm	
R $\downarrow$ = 2.6 μm	
Max Reversal Error:	
B = 4.1 μm	
Mean Reversal Error:	
$\bar{B}$ = -3.7 μm	
Raw Accuracy:	
Araw = 8.9 μm	
Araw $\uparrow$ = 5.3 μm	
Araw $\downarrow$ = 5.0 μm	
Raw Repeatability:	
Rraw = 5.5 μm	
Rraw $\uparrow$ = 1.9 μm	
Rraw $\downarrow$ = 1.6 μm	
6 Sigma (total pop.):	
6 σ = 14.7 μm	
Slope (End Points):	
Slope = -1.5 $\mu\text{m}/\text{m}$	
Slope (Least Squares):	
Slope = -2.4 $\mu\text{m}/\text{m}$	

Data Analysis	
Standards: VDI 3441/2617	
Accuracy:	
P = 10.6 μm	
A $\uparrow$ = 7.3 μm	
A $\downarrow$ = 6.7 μm	
P σ max = 4.0 μm	
$\bar{P}_s$ = 2.8 μm	
Mean Bidir. Pos. Dev.:	
Pa = 4.0 μm	
Accuracy:	
Ps+U = 7.9 μm	
Max Reversal Error:	
Umax = 4.1 μm	
Mean Reversal Error:	
$\bar{U}$ = 3.7 μm	

E1733A Automation Interface Operation

Left Column Item	'CI_* Constant	Right Column Item	'CI_* Constant
Accuracy Overall	BIACCURACY	VDI Pos. Accuracy	VDI_P
Accuracy Fwd	FWDACCURACY	VDI Fwd Accuracy	FWDACCURACY
Accuracy Rev	REVACCURACY	VDI Rev Accuracy	REVACCURACY
Sys Dev Overall	BISYSPOSDEV	VDI PS max	VDI_PSMAX
Sys Dev Fwd	FWDSYSPOSDEV	VDI PS mean	VDI_PSMEAN
Sys Dev Rev	REVSYSPOSDEV	VDI BiDir Pos. Acc.	VDI_PA
Mean BiDir Pos. Dev	BIMEANDEV	VDI Ps+U	VDI_PSU
Repeat Overall	BIREPEATPOS	VDI Max Rev Err	VDI_UMAX
Repeat Fwd	FWDREPEATPOS	VDI Mean Rev Err	VDI_UMEAN
Repeat Rev	REVREPEATPOS		
Max Rev Error	BIMAXREV		
Mean Rev Error	BIMEANREV		
Raw Accuracy Overall	BIRAWACC		
Raw Accuracy Fwd	FWDRAWACC		
Raw Accuracy Rev	REVRAWACC		
Raw Repeat Overall	BIRAWREP		
Raw Repeat Fwd	FWDRAWREP		
Raw Repeat Rev	REVRAWREP		
6 Sigma	SIXSIGMA		
Slope End Points	SLOPEEP		
Slope Least Squares	SLOPELS		

Flatness and Way Straightness:

Data Analysis	Axis:All
Moody	
Max. Elevation: 178.5 μin	
Plus/Minus: 89.3 μin	
Closure #7(DH): 1.5 μin	
Closure #8(BF): 15.0 μin	

Data Analysis
Max. Elevation: 2.248 μm

Item	'CI_* Constant	Notes
Analysis Method	--	Not available
Max Elevation	'CI_MAX_ELEVATION	
Plus/Minus	'CI_PLUSMINUS	
DH Closure	'CI_CLOSURE_DH	
BF Closure	'CI_CLOSURE_BF	

E1733A Automation Interface Operation

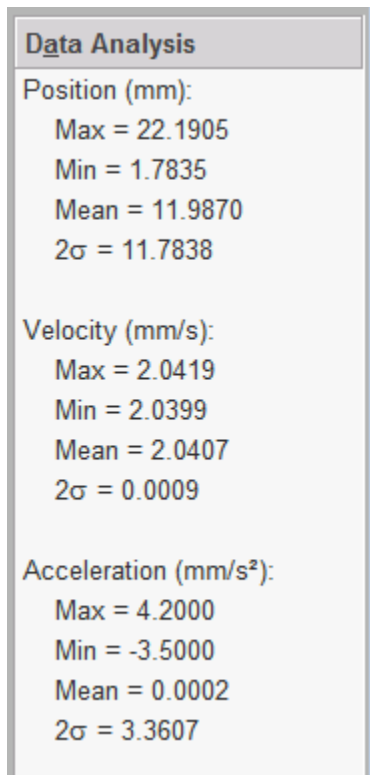
Parallelism and Squareness:

Data Analysis	Axis:All	Data Analysis	Axis:All
Standards: ISO 230-2 1997 (M)		Standards: ISO 230-2 1997 (M)	
Parallelism = -1.02 arcsec		Squareness = 0.00 arcsec	
(1) Axis: X	(2) Axis: Y	(1) Axis: X	(2) Axis: Y
(1) Accuracy:	(2) Accuracy:	(1) Accuracy:	(2) Accuracy:
A1 = 5.6 μ m	A2 = 5.8 μ m	A1 =	A2 =
A1 $\uparrow$ = 4.6 μ m	A2 $\uparrow$ = 5.0 μ m	A1 $\uparrow$ =	A2 $\uparrow$ =
A1 $\downarrow$ = 4.9 μ m	A2 $\downarrow$ = 5.2 μ m	A1 $\downarrow$ =	A2 $\downarrow$ =
(1) Sys. Dev. Pos.:	(2) Sys. Dev. Pos.:	(1) Sys. Dev. Pos.:	(2) Sys. Dev. Pos.:
E1 = 5.6 μ m	E2 = 5.8 μ m	E1 =	E2 =
E1 $\uparrow$ = 4.6 μ m	E2 $\uparrow$ = 5.0 μ m	E1 $\uparrow$ =	E2 $\uparrow$ =
E1 $\downarrow$ = 4.9 μ m	E2 $\downarrow$ = 5.2 μ m	E1 $\downarrow$ =	E2 $\downarrow$ =
(1) Mean Bidir. Pos. Dev.:	(2) Mean Bidir. Pos. Dev.:	(1) Mean Bidir. Pos. Dev.:	(2) Mean Bidir. Pos. Dev.:
M1 = 4.8 μ m	M2 = 5.1 μ m	M1 =	M2 =
(1) Repeatability:	(2) Repeatability:	(1) Repeatability:	(2) Repeatability:
R1 = 1.1 μ m	R2 = 1.2 μ m	R1 =	R2 =
(1) Max Reversal Error:	(2) Max Reversal Error:	(1) Max Reversal Error:	(2) Max Reversal Error:
B1 = 1.1 μ m	B2 = 1.2 μ m	B1 =	B2 =
(1) Mean Reversal Error:	(2) Mean Reversal Error:	(1) Mean Reversal Error:	(2) Mean Reversal Error:
$\bar{B}1$ = -1.0 μ m	$\bar{B}2$ = -1.0 μ m	$\bar{B}1$ =	$\bar{B}2$ =
(1) Raw Accuracy:	(2) Raw Accuracy:	(1) Raw Accuracy:	(2) Raw Accuracy:
Araw1 = 5.6 μ m	Araw2 = 5.8 μ m	Araw1 =	Araw2 =
Araw1 $\uparrow$ = 4.6 μ m	Araw2 $\uparrow$ = 5.0 μ m	Araw1 $\uparrow$ =	Araw2 $\uparrow$ =
Araw1 $\downarrow$ = 4.9 μ m	Araw2 $\downarrow$ = 5.2 μ m	Araw1 $\downarrow$ =	Araw2 $\downarrow$ =
(1) Raw Repeatability:	(2) Raw Repeatability:	(1) Raw Repeatability:	(2) Raw Repeatability:
Rraw1 = 1.1 μ m	Rraw2 = 1.2 μ m	Rraw1 =	Rraw2 =
(1) 6 Sigma (total pop.):	(2) 6 Sigma (total pop.):	(1) 6 Sigma (total pop.):	(2) 6 Sigma (total pop.):
6 σ 1 = 3.1 μ m	6 σ 2 = 3.2 μ m	6 σ 1 =	6 σ 2 =
(1) Slope (Least Squares):	(2) Slope (Least Squares):	(1) Slope (Least Squares):	(2) Slope (Least Squares):
Slope1 = 97.2 μ m/m	Slope2 = -107.1 μ m/m	Slope1 =	Slope2 =
(1) Slope (End Points):	(2) Slope (End Points):	(1) Slope (End Points):	(2) Slope (End Points):
Slope1 = 95.8 μ m/m	Slope2 = -102.4 μ m/m	Slope1 =	Slope2 =

Item	CI * Constant	Notes
Parallelism	PAR_RESULT	
Squareness	SQU_RESULT	
Axis 1 Values	See Linear side of Linear and VDI	Set NUM value to 1
Axis 2 Values	table above	Set NUM value to 2

E1733A Automation Interface Operation

Time Base:



Item	'CI_ * Constant
Position Max	POS_MAX
Position Min	POS_MIN
Position Mean	POS_MEAN
Position 2Sigma	POS_NSIGMA
Velocity Max	VEL_MAX
Velocity Min	VEL_MIN
Velocity Mean	VEL_MEAN
Velocity 2Sigma	VEL_NSIGMA
Acceleration Max	ACC_MAX
Acceleration Min	ACC_MIN
Acceleration Mean	ACC_MEAN
Acceleration 2Sigma	ACC_NSIGMA

E1733A Automation Interface Operation

Read Status and Data Info ('CC_READ*):

The automation interface also allows one to read the program status and measurement data using the 'CC_READ* commands and 'CI_* index values in the following table. See the header file for details.

Item for 'CC_READSTATUS Command	'CI_* Index Constant	Item for 'CC_READERRORDATA Command	'CI_* Index Constant
What type of window is active?	ACTIVEWINDOW	All measurements axis 1 data	ERRAXIS1
Is laser head producing a reference signal?	5519ABREADY	Parallelism, Squareness, and Flatness axis 2 data	ERRAXIS2
Is E1735A detected/ready?	E1735AREADY	Flatness axis 3 data	ERRAXIS3
How many E1735A are connected?	E1735ACOUNT	Flatness axis 4 data	ERRAXIS4
Is E1736A detected/ready?	E1736AREADY	Flatness axis 5 data	ERRAXIS5
How many E1736A are connected?	E1736ACOUNT	Flatness axis 6 data	ERRAXIS6
How many E1737A are connected?	E1737ACOUNT	Flatness axis 7 data	ERRAXIS7
How many E1738A are connected?	E1738ACOUNT	Flatness axis 8 data	ERRAXIS8
Is a 55290B Table detected/ready?	55290BREADY		
What is the beam strength %?	BEAMSTRENGTH		
Is the current setup ok?	MEASURESETUPOK		
Is an error message displayed?	HASERRORMESSAGE		
What is the error message string?	ERRORMESSAGE		
Is a measurement in process?	ISMEASURING		
How many samples are collected?	SAMPLECOUNT		
Is the measurement complete?	ACCOMPLISHED		
What is value in the upper display?	UPPERDISPLAY		
What is value in the lower display?	LOWERDISPLAY		
How many places used in upper?	UPPERSELECTION		
How many places used in lower?	LOWERSELECTION		
What are upper display units?	UPPERUNITS		
What are lower display units?	LOWERUNITS		
What is upper display string?	UPPERSTRING		
What is lower display string?	LOWERSTRING		
What is current cycle number?	CURRENTCYCLE		
What is current axis?	CURRENTAXIS		
What is measurement direction?	NEXTDIRECTION		
What is next index (point)?	NEXTINDEX		
What is target position?	NEXTTARGET		